



An experimental and modeling study of ethylene–air combustion over a wide temperature range

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ABSTRACT

Ethylene is an important intermediate during oxidation and pyrolysis process of higher hydrocarbons, and also serves as a fuel under practical engines. An experimental and modeling study is performed to investigate the combustion chemistry of ethylene–air mixtures under a wide range of conditions. The shock tube experiments are conducted at pressures of 1, 4, 10 and 19 atm with equivalence ratios of 0.5, 1.0 and 2.0 to determine the ignition delay times by measuring the emission signal of excited methylidyne (CH^*). The ignition data under 10 and 19 atm both span a wide temperature range varying from a low temperature of 721 K to a high temperature of 1320 K. No typical negative-temperature-coefficient is observed, but the ignition at low temperatures is much shorter than the extrapolation at high temperatures. Simulations are performed and compared to the experiments using contemporary core models with consideration of a linear pressure rise of 3% /ms, but no simulation could predict the ignition trend under low-to-intermediate temperatures. Therefore, a detailed model of ethylene oxidation is proposed based on AramcoMech2.0 in the present work.

The modeling behavior at high temperatures is improved by revision of addition reaction between ethylene and oxygen atom ($\text{C}_2\text{H}_4 + \text{O}$). The modeling behavior at the region of low-to-intermediate temperatures is improved by detailed updating of β -hydroxyethyl oxidation ($\text{PC}_2\text{H}_4\text{OH} + \text{O}_2$). The effect of reactions at the potential energy surface of $\text{C}_2\text{H}_4 + \text{HO}_2$ and $\text{C}_2\text{H}_5 + \text{O}_2$ is determined, and the second oxygen addition ($\text{C}_2\text{H}_4\text{OOH} + \text{O}_2$) is added into current model. The present model is extensively validated by ignition delay times, concentration profiles of stable species in jet-stirred reactors and laminar flame speed under various test conditions. Sensitivity analyses are carried out to identify the dominant reactions during the auto-ignition process and laminar premixed flame propagation. Additional simulations are performed to investigate the effect of model modifications on predictions of ethane and ethanol oxidation. The present model slightly promotes the low-temperature ignition of both ethane and ethanol, and its prediction on the concentration profiles of some stable species is also improved. The proposed model of ethylene oxidation can be used as a cornerstone to develop models of larger hydrocarbon fuels.

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1. Introduction

As the smallest alkene, ethylene (C_2H_4) is an important intermediate species in the combustion or pyrolysis of larger hydrocarbons [1,2], as well as atmospheric chemistry, and is mainly formed by the β -scission reaction of alkyl radicals during

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the reaction of almost all fuels. Previous studies [3,4] has revealed the important role played by ethylene during the oxidation of n-alkanes because the reactions concerned with ethylene obviously influence the combustion parameters, such as the ignition delay time and laminar flame speed. Moreover, due to the greater reaction rate, burning velocities and detonability compared with most other hydrocarbon fuels, ethylene can serve as a fuel of interest in a variety of practical engines, including turbines and scramjet [5], and can be applied to hypersonic propulsion [6]. Furthermore, ethylene is an important precursor of poly-aromatic hydrocarbons (PAH) and soot particles [7,8], whose impact on health and the

Table 1

The available experimental data for ethylene oxidation on ignition delay times from literature.

Reactor	Mixture	<i>T</i> (K)	<i>p</i> (atm)	ϕ	Dilution%	Diag. ^a	Ref.
ST	C ₂ H ₄ /O ₂ /Ar	1090–1520	0.059–6.3	0.33–3	61.9%–99.3%	dp/dt	[16]
ST	C ₂ H ₄ /O ₂ /Ar	1000–1650	2–18	1–3	93%–98%	CH ⁺ /OH ⁺	[17]
ST	C ₂ H ₄ /O ₂ /Ar	1092–1743	1.3–3.0	1	75%–96%	CH ⁺	[19]
ST	C ₂ H ₄ /O ₂ /N ₂	1060–1520	5.9–16.5	0.5–2	Air ^b	CH ⁺ /OH ⁺ /C ₂	[5]
ST	C ₂ H ₄ /O ₂ /N ₂	1003–1401	1.1–24.9	0.3–2	Air ^b	OH ⁺	[20]
ST	C ₂ H ₄ /O ₂ /N ₂	800–1620	0.053–0.105	1–1.5	50%–75%	CH ⁺	[21]
RCM	C ₂ H ₄ /O ₂ /N ₂ /Ar	850–1050	15–50	1	89.43% ^c	dp/dt	[22]
ST/RCM	C ₂ H ₄ /O ₂ /N ₂ /Ar	770–1515	20–40	0.5–2	75%–90% ^d	dp/dt	[23]

^a Diag. denotes the diagnostic method with pressure or CH⁺/OH⁺ chemiluminescence signals to determine the ignition delay time.^b N₂/O₂ = 3.76.^c Including 15.86% of N₂ and 73.57% of Ar.^d 90% indicates 90% of N₂ dilution, or 45% of N₂ and 45% of Ar, respectively.

environment has been elucidated, during the processes of pyrolysis and premixed flame [9–11]. Due to the important role of ethylene in the above aspects, a better understanding of the combustion chemistry of ethylene is a prerequisite for developing the chemical kinetic models of larger hydrocarbons and even oxygenated fuels [12,13] improving the efficiency of practical engines and reducing pollutant formation.

Autoignition is a spontaneous process whereby a fuel/air mixture undergoes a complex chemical process leading to combustion without the aid of an external source [14]. Ignition delay time serves as a tool not only to control the performance and emission of fuel, but also to validate the reliability of chemical kinetics model. The characteristic of low-temperature ignition of fuel is the basis of octane rating system for gasoline and the cetane rating system for diesel, and is particularly critical in the internal combustion engines [15]. The experimental data of ignition delay times for ethylene oxidation, available from literature up to now, are summarized in Table 1 in which the abbreviations ST and RCM stand for shock tube and rapid compression machine, respectively. Regarding the ignition properties of ethylene, many recent studies focused on ethylene oxidation in argon-diluted (Ar-diluted) atmosphere behind reflected shock waves under a high-temperature range. Tereza et al. [16] studied the ignition delay time of C₂H₄/O₂/Ar mixtures at temperatures of 1090–1250 K and pressure of 6.5–atm and developed a kinetic model to describe the characteristics of autoignition at high temperatures. An ignition delay study under different Ar-diluent conditions of 93%, 96% and 98% vol, temperatures of 1000–1650 K, and pressures (*p*) of 2, 10, 18–atm, at equivalence ratios (ϕ) of 1 and 3 was described in detail by Saxena et al. [17]; in this study, USC model version II [18] developed by Wang et al. could be used to reestablish the ignition characteristics under stoichiometric, fuel-rich and low pressure conditions, but the experimental data could not be predicted at a lower temperature of approximately 1000 K and high pressures of 10–atm and 18–atm. Xiong et al. [19] measured the ignition delay time of C₂H₄/O₂/Ar stoichiometric mixtures at 1092–1743 K and pressures of 1.3–3.0–atm under diluent mole fractions of 96% and 75% with shock tube. They concluded that the diluent rate had no impact on the reaction paths of ethylene oxidation. Moreover, only limited studies concentrated on the combustion of ethylene under real fuel-air conditions. Penyazkov et al. [5] investigated the oxidation of ethylene/air mixtures at a temperature range of 1060–1520 K, pressures of 5.9–16.5–atm, and equivalence ratios of 0.5, 1.0 and 2.0. Shock-tube measurements of ethylene/air mixtures have also been performed by Kopp et al. [20] in the temperature range of 1003–1401 K over a wide range of equivalence ratios (0.3–2.0) and pressures (1.1–24.9–atm). They observed no pressure effect on fuel-lean mixtures, while a decreasing ignition delay time was observed with increasing pressure under stoichiometric and fuel-rich conditions. Under low-temperature conditions, there are

few experiments studying ignition characteristics of ethylene oxidation. Cadman et al. [21] studied ethylene/O₂/N₂ combustion at equivalence ratios of 1 and 1.5 over a wide range of temperatures (800–1620 K) behind reflected shock wave and found that the ignition delay times at lower temperatures (<1000 K) were much shorter than extrapolation from high temperature data. Kumar et al. [22] measured the ignition delay time of C₂H₄/O₂/N₂/Ar diluent mixtures using a rapid compression machine (RCM) at compressed pressures from 15 to 50 bar and the temperatures ranging from 850 to 1050 K, and found that the reactions among ethylene and hydroperoxyl radical (C₂H₄+H₂O₂), vinyl radical and oxygen molecule (C₂H₃+O₂) were sensitive to ignition. Furthermore, they concluded that significant improvements in the model should be made at low temperatures and high pressures. Most recently a comprehensive study aimed at ignition delay times of prominent C₁–C₂ hydrocarbons including methane, ethane, and ethylene was performed by the group of Curran [23]. This research involved conditions over a wide range of temperatures (~800–2000 K), pressures (~1–80 bar), equivalence ratios (~0.5–2.0) and dilutions (~75–90%), which were obtained by ST and RCM experiments not only in their study [23], but also other available data from literature. Although the presented NUIGMech0.9 can suitably predict the measured ignition delay times over the wide range of conditions, some deviations are observed at low-temperature end. Currently, due to the limitation of the scope of experimental data, more experiments concerning the ignition delay time should be performed to describe the real process of combustion and identify a chemical kinetic model suitable for a wide range of conditions.

Regarding chemical kinetic models simulating ethylene oxidation, several popular chemical models can be used as candidates, as totally summarized in Table 2. The LLNL n-butane model [24] was validated in flames of methane (CH₄), ethane (C₂H₆), C₂H₄ and propane (C₃H₈). Konnov presented a well-validated model of the combustion of small hydrocarbons [25]. A small detailed model named UCSD involving 50 species and 247 elementary reactions was updated by Prince and Williams [26], who investigated the low-temperature ignition of C₃H₈ and C₂H₆. Glarborg et al. [27] developed a model of the oxidation of C₂H₄, which has been well validated by a high-pressure flow reactor at 600–900 K and 60 bar. A kinetic model of the high-temperature ignition of ethylene based on shock wave measurements was constructed by Tereza et al. [16] Wang et al. [18] developed a high-temperature USC model version II (marked as USC 2.0 in the following part) of the combustion of H₂/C₀/C₁–C₄ compounds. Ranzi et al. [28,29] proposed the so-called Creck model by lumping high and low temperature reactions, which is applicable to pyrolysis, partial oxidation and combustion of hydrocarbons for C₁–C₁₆ species. Zhou et al. developed a detailed model of AramcoMech2.0 including the sub-mechanisms of isobutene [30], 1-butene [31], 2-butene [32], propene [33,34] and ethylene [35]. Furthermore,

Table 2

The summary of popular core-mechanisms aimed for smaller hydrocarbons, which can be employed to simulate ethylene oxidation.

Model	Composition	Temperature range	No. of species	No. of reactions	Ref.
LLNL	H ₂ /C ₁ –C ₄	High Temperature	155	689	[24]
Konnov's	H ₂ /C ₁ –C ₄	High Temperature	129	1231	[25]
UCSD	H ₂ /C ₁ –C ₃	Wide Range ^a	50	247	[26]
Glarborg's	H ₂ /C ₁ –C ₃	Wide Range	116	962	[27]
USC II	H ₂ /C ₁ –C ₄	High Temperature	111	784	[18]
AramcoMech2.0	H ₂ /C ₁ –C ₄	Wide Range	494	2716	[30]
Creck	H ₂ /C ₁ –C ₁₆	Wide Range	451	17,747	[29]

^a Including both high and low-temperature chemistry.

Xu and Konnov [36] compared the simulation results of different contemporary models, including Konnov's, LLNL, UCSD and USC, of ethylene ignition with experimental data at high temperatures. The simulation of the ignition delay time and laminar flame speed showed noticeable differences by employing different models. The reactions involving ethylene and vinyl chemistry were considered as the main reason leading to the different modeling behaviors and should be reconsidered in detail in the future [37].

Given the above description of C₂H₄ oxidation, there is a lack of ignition data under practical C₂H₄/air conditions, especially at low temperatures and high pressures, which is more relevant to real engines. In the present work, the combustion mechanism for C₂H₄/air mixture is elucidated through experimental and modeling studies. Systematic experiments are performed to determine the ignition delay time for C₂H₄/air mixtures at a wide range of temperatures from 721–1320 K, pressures from 1–atm to 19–atm, and equivalence ratios of 0.5–2. As discussed in Section 3, due to the significant deviation among the current chemistry kinetic models, especially under intermediate-to-low temperatures (≤ 1000 K), a detailed model of C₂H₄/air oxidation is developed based on AramcoMech2.0 [30] to describe the autoignition characteristics of C₂H₄ over a wide range of real-engine conditions. This model is further validated with experimental data of species profiles in JSR [38,39] and laminar flame speed [22,40–45] at various pressures, temperatures and different dilution ratios. Sensitivity analyses are carried out to determine the dominant reactions under the autoignition process and laminar premixed flame propagation. The proposed model is helpful for improving the combustion model of larger hydrocarbons and developing low-temperature combustion engine technologies.

2. Experimental and simulation method

Experiments of ignition delay times are carried out in a stainless-steel shock tube. This facility has been described in detail and successfully used in our previous works aimed to study the ignition from low to high temperature range [46–49], thus only a brief description is given in this section. The shock tube comprises a 6 m driver section and a 5 m driven section with an inner diameter of 10 cm separated by a double diaphragm section. Helium is used as the driver gas at a high temperature where the ignition delay time is less than 2 ms, and for a longer time measurement, tailored helium/nitrogen driver gas mixtures are used to extend the test times up to 15 ms. The detection limit of ST is determined by the work time, and the work time of 15 ms by tailored technique leads to a limited ignition temperature of 700 K to our experiments. The gas mixtures of ethylene and synthetic air are prepared manometrically in a 40 L mixture tank and mixed at least 3 h prior to experiments.

A series of four piezoelectric pressure sensors (PCB 113B) installed along the last 1 m of the driven section are used to measure the incident shock-wave velocities. A linear extrapolation of the measured velocities is used to determine the incident wave speed at the end wall location. The end wall velocity, the pressure

and gas composition of the test gas are used to calculate the reflected-shock pressure and temperature by one-dimensional shock relations. A grating monochromator coupled with a photomultiplier tube is used to detect the excited methylidyne (CH*) chemiluminescence at 431 nm at the same cross section as the last pressure transducer is 15 mm away from the shock tube end wall. The ignition delay time is defined as the time interval between the arrival of the reflected shock wave indicated by the jump of pressure signal and the onset of ignition determined by extrapolating the maximum slope of CH* emission signal back to the baseline. The sharp rising of pressure and high intensity of radical emission make the ignition determination clear, and an example of ignition delay measurements at typical high and low temperature is shown in Fig. 1. The non-ideal boundary layer effect and fluid dynamic nonidealities can cause obvious pre-ignition energy release, especially for high fuel-concentration system without argon-dilution in this work, which may lead to dramatic pre-ignition pressure rise [50,51]. This issue has been enunciated in previous researches [52–54], and will lead to much lower ignition delay times at low temperatures. Therefore, the CONP (i.e., constant enthalpy and pressure) modeling code coupled with VITM code (i.e., volume as a function of time) is employed to explain the facility effect in present work [53]. This method has been extensively applied to various shock-tube modeling studies [46,49,52,53], and great agreements were achieved by comparing against experimental data. In this work, the variation in the pressure is determined to be in the range of $(dp/dt)/(1/p_0) = 1\text{--}5\%$ /ms, and a typical pressure gradient with the linear relation of $(dp/dt)/(1/p_0) = 3\%$ /ms is used in shock tube simulations in our work. The data of pressure history and CH* trace at representative conditions are given in Supplementary material as references.

All of the simulations in this work are performed with the Chemkin-pro package [55] to obtain the ignition delay time, species concentration profile and laminar flame speed. To test the influence of facility pressure-effect, a simulation for ethylene/air oxidation at a high-pressure condition of 10 atm is performed using AramcoMech2.0 [30], as shown in Fig. 2. Very little difference is observed by using traditional adiabatic CONV code (i.e., constant internal energy and volume) and VITM code at relatively short measurement time ≤ 2 ms. However, great disparity arises at longer measurement time for low-temperature ignition. Thus, facility effect in this work must be considered for all of the modeling study at measurement time > 2 ms, and will not be emphasized again in the following discussions. Simulations of ST are performed with zero-dimensional batch reactor module embedded in Chemkin-pro package [55] and begin at the onset of the reflected shock period. Constant volume, homogeneous, and adiabatic assumptions (CONV) are carried out for calculations of high-temperature conditions. The simulated ignition delay times are defined in line with the diagnostic method of shock-tube measurement. However, when certain simulation result of CONV is more than 2 ms, the CONP coupled with VITM code using a pressure rise of 3%/ms as presented above is performed at this condition. The volume profiles converted from pressure profiles

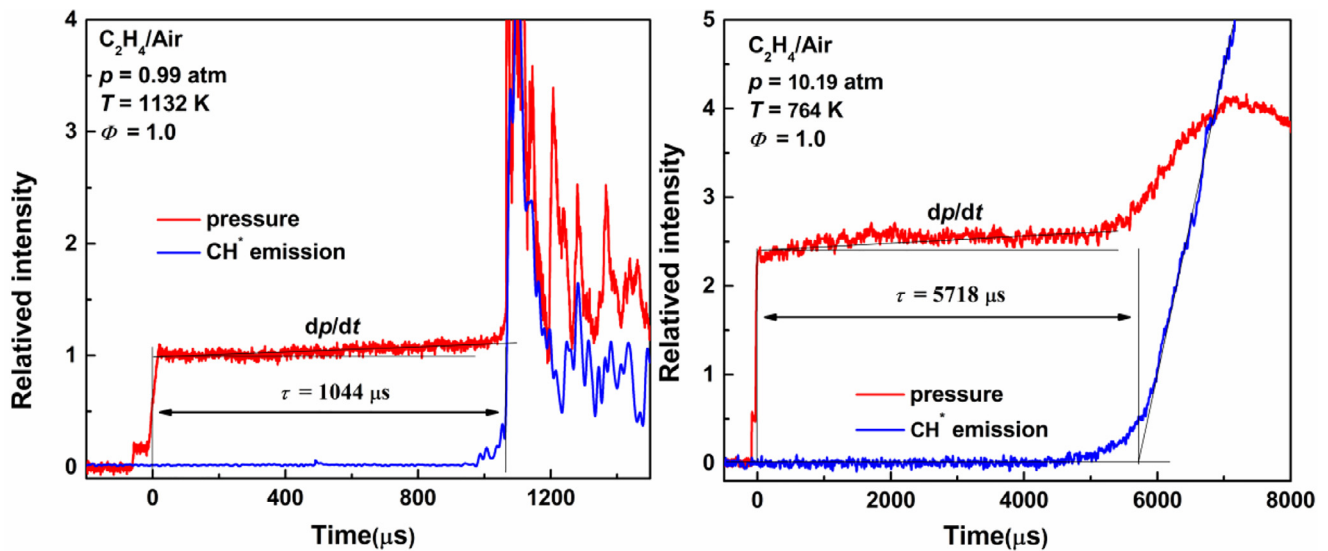


Fig. 1. Typical pressure and CH* signals in ignition delay time measurement.

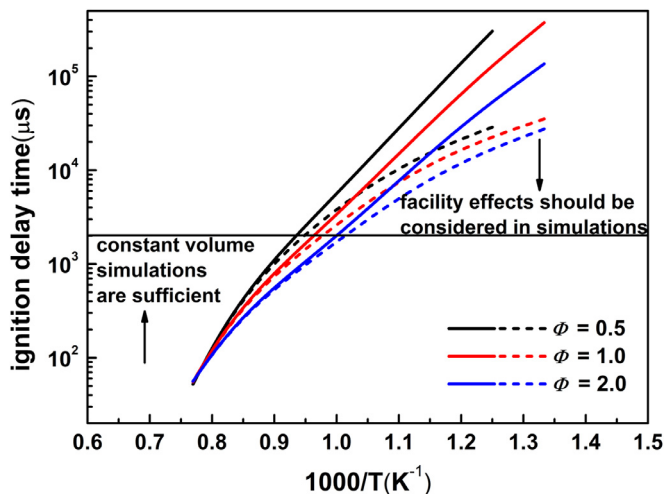


Fig. 2. Simulation results employing AramcoMech2.0 [30] for ethylene/air mixtures at 10 atm. Solid lines indicate adiabatic constant volume simulations while dashed lines represents VITM method with a linear pressure rise of 3%/ms.

using isentropic equation $(p_2/p_1) = (V_1/V_2)^\gamma$ are used as input parameters of VPRO keyword in Chemkin-pro [55]. Overall uncertainty in the measured ignition delay time is estimated within $\pm 20\%$ combining all of the uncertainties from the calculated temperature and pressure, the composition of the ethylene/air mixture, and the ignition delay determination from the pressure and CH* signals. The specific method of uncertainty analysis is presented in Supplementary material. The ignition delay time data obtained in this work are given in Table S1 of Supplementary material. Further comparisons between our experimental measurements and the results of Kopp et al. [20] at same conditions are given in Fig. S3 to prove the rationality of our data.

Flux analyses are carried out to determine the most dominant reactions responsible for fuel consumption at typical conditions. All of the flux analyses presented in this work are carried out by rate of production (ROP) analyses assuming a constant volume reactor with Chemkin-pro software package [55], and the reaction time when fuel is 20% consumed is chosen as a representative condition. The corresponding flux ratios are normalized ROP for a specific species which serve as an important intermediate in the process of oxidation.

JSR simulations are performed by Chemkin-pro [55] using perfectly stirred reactor module with the transient solver and an end time of 15 s.

The simulations of laminar flame speed are performed with the Premix module in Chemkin-pro [55]. Simulations are converged to a grid-independent solution of approximately 1000 grid points, meanwhile GRAD and CURV values of 0.05 were assigned. For calculating the transport coefficients, the mixture-average model is used. Furthermore, the effect of thermal diffusion (Soret effect) is considered in our calculations. The sensitivity analyses of flame speed are performed with inbuilt utility of Chemkin-pro, which calculates the first-order sensitivity coefficients for the predicted mass flow rate. A positive value indicates increasing reactivity, and vice versa.

3. Model development

The modeling results of C₂H₄/air ignition obtained with five contemporary core models of small hydrocarbons [18,26,27,29,30] are compared with the shock tube measurements presented in this work. Figure 3 shows the comparison at a pressure of 10-atm and various equivalence ratios ranging from 0.5 to 2.0. Great deviations can be observed for all above models, especially at the low-intermediate temperature range, suggesting that the chemical kinetic model of C₂H₄ oxidation should be improved even if the facility effect has been taken into account. Such improvement could provide a further understanding of the reaction characteristic of C₂H₄ oxidation and serve as a guide in the design of practical engines under low-to-high temperature ranges.

Recently, many studies have presented detailed oxidation pathways for higher alkenes, such as propene, butene and even hexane [30–34,56,57]. However, for ethylene, the combustion reactions differ from those of the higher alkenes due to the lack of the formation of alkenyl and allylic radicals during the process of oxidation. Considering both the commonalities and differences between ethylene and the higher alkenes in recent studies, a detailed kinetic model based on the generalized reaction scheme of ethylene combustion with 149 species and 567 reactions is developed in this work. The main framework of the model is given in Fig. 4. Almost all of the dominant reactions are initiated from ethylene, vinyl radical generated by H abstraction, and ethyl radical from H addition reaction. The three species are marked as F, F' and R, respectively. To ensure the completeness of reaction

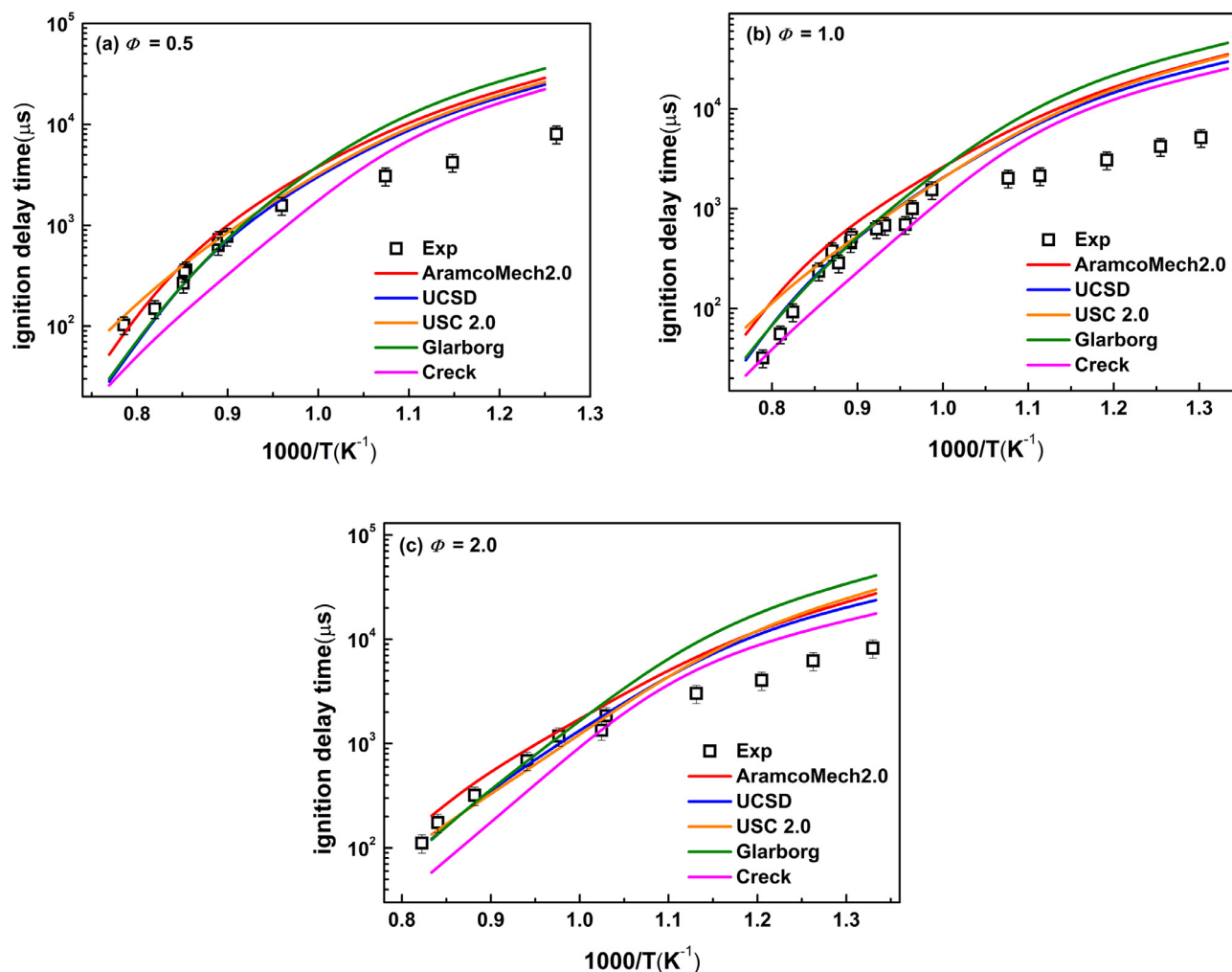


Fig. 3. Comparison of our experimental data with predictions using contemporary models at 10 atm with (a) $\phi = 0.5$, (b) $\phi = 1.0$ and (c) $\phi = 2.0$. Symbols: experimental data; Lines: simulations.

networks of C_0 – C_2 sub-mechanism, AramcoMech2.0 [30] which aimed at various individual and hybrid fuels, is chosen as a foundation for the model development in this work. Some of updates and additions for C_2 chemistry based on Fig. 4 are carried out to improve the modeling performance for ethylene oxidation, as summarized in Table S2 of Supplementary material, which will be discussed in detail in this section. The thermodynamic data of the species absent from AramcoMech2.0 are obtained by the THERM program [58], and the lacking transport data are supplied by the method described by Wang and Frenklach [59]. The kinetic, thermodynamic and transport files used for Chemkin-pro simulation are provided in Supplementary material. The chemical structures and abbreviations of all species presented in this work are provided in Scheme S1 of Supplementary material.

3.1. $C_2H_4 + \dot{O}$ addition reaction

- $C_2H_4 + \dot{O} = \dot{C}H_3 + H\dot{C}O$ (R1–1)
- $C_2H_4 + \dot{O} = \dot{C}H_2CHO + \dot{H}$ (R1–2)
- $C_2H_4 + \dot{O} = CH_2(X^3B_1) + CH_2O$ (R1–3)
- $C_2H_4 + \dot{O} = CH_3\dot{C}O + \dot{H}$ (R1–4)
- $C_2H_4 + \dot{O} = CH_2CO + H_2$ (R1–5)

In the present work, we find that this reaction class is sensitive to both autoignition and laminar premixed flame propagation and becomes more important as the temperature increases. From

the first reaction of R1–1 to the fifth reaction of R1–5, they are chemical activation reactions via triplet adduct $\dot{C}H_2CH_2\dot{O}$. Baulch et al. [60] recommended three pathways (R1–1, R1–2 and R1–5) with estimated branching ratios of 0.35, 0.6 and 0.05 over the temperature range of 220 – 2000 K. Nguyen et al. [61] used various quantum chemical methods to investigate the potential energy surface of the $C_2H_4 + \dot{O}$ reaction within intersystem crossing (ISC) of the triplet adduct $\dot{C}H_2CH_2\dot{O}$ to the singlet surface. They predicted that the yields at 300/2000 K are $\dot{C}H_3 + H\dot{C}O$ (0.48/0.37), $\dot{C}H_2CHO + \dot{H}$ (0.40/0.19) and $CH_2(X^3B_1) + CH_2O$ (0.05/0.29). AramcoMech2.0 only considered the pathways of R1–1 and R1–2 and modified the branching ratios to 0.55/0.45 based on the overall rate constants provided by Baulch et al. [60]. In the present work, we consider five pathways (from R1–1 to R1–5) in the developed model, and the corresponding kinetic parameters are adopted from the latest theoretical research with a high accuracy by Li et al. [62] based on the master equation approach, including a priori description of ISC. They predicted that $\dot{C}H_2CHO + \dot{H}$ and $CH_2(X^3B_1) + CH_2O$ are the dominant products at elevated temperatures. The results were different from the previous conclusions reported by Baulch and Nguyen [60,61], who considered $\dot{C}H_3 + H\dot{C}O$ as the major products at both room and high temperatures. It is worth noting that pressure dependence and stabilization of intermediates are negligible under conditions relevant to combustion as noted in the theoretical analysis of Li et al. [62]. Therefore, the

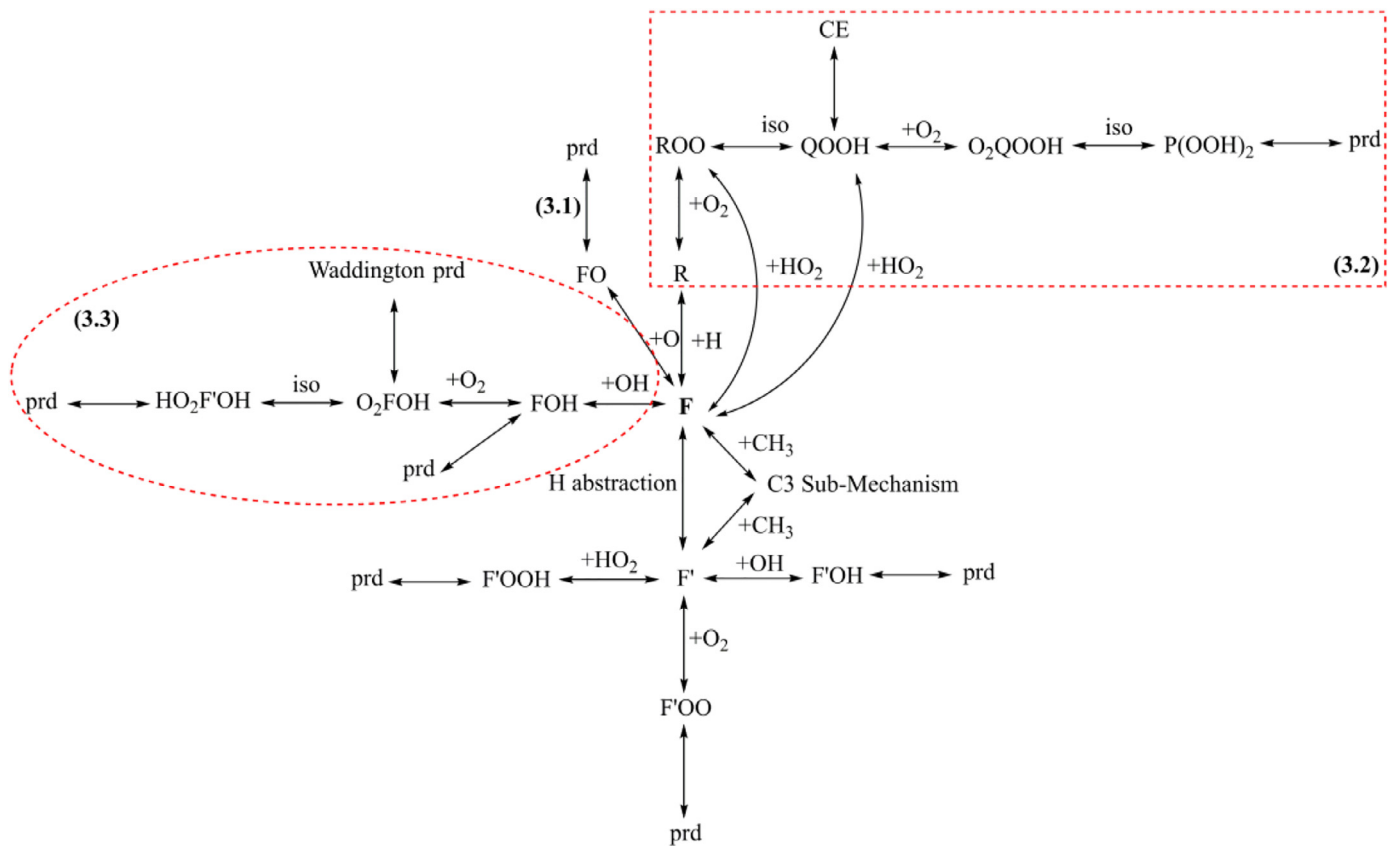


Fig. 4. Generalized reaction pathways of ethylene oxidation considered in this work. F: fuel, referring to ethylene; F': vinyl radical; R: ethyl radical; iso: isomerization reactions; CE: cyclic ether; KHP: ketohydroperoxide; prd: oxygenated products or intermediates. The corresponding reaction classes discussed below are highlighted in bold numbers.

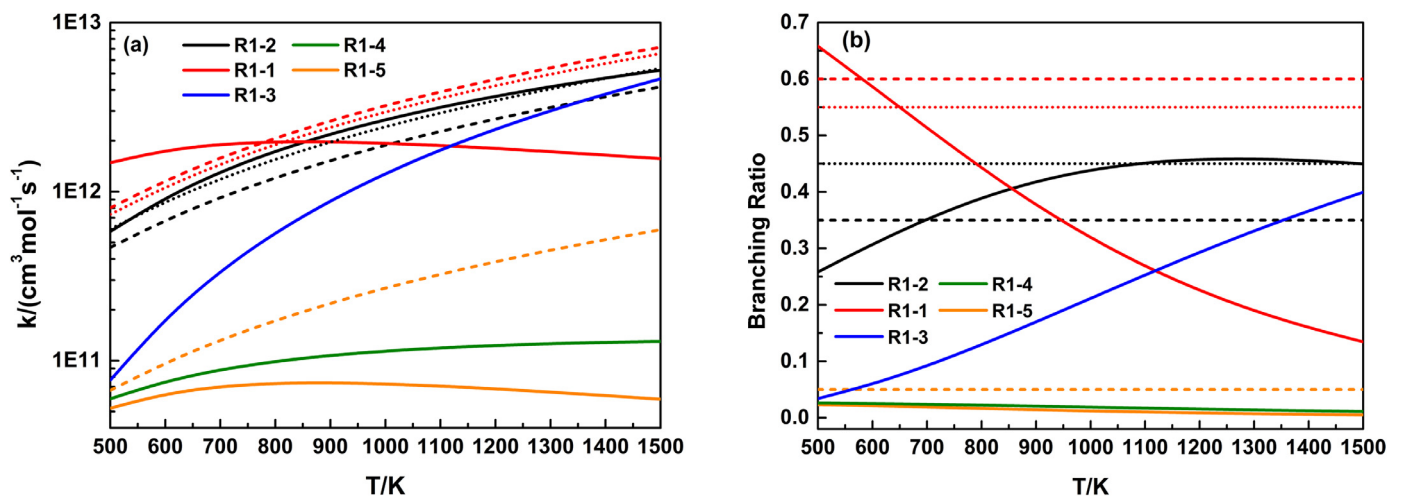


Fig. 5. Rate constants (a) and branching ratios (b) of the major pathways of the addition reaction between C_2H_4 and $\dot{O}$ over a temperature range of 500 - 1500 K. Solid lines: present model based on the theoretical investigation by Li et al. [62]; Dashed lines: recommendation by Baulch et al. [60]; Dotted lines: AramcoMech2.0 [30].

pressure-dependent kinetics and stabilization pathways forming C_2H_4O1-2 (oxirane), CH_3CHO , CH_2CHOH , $CH_2CH_2\dot{O}$ and so on are not considered. In addition, all singlet species, such as CH_3CHO , CH_2CHOH , C_2H_4O1-2 , are not dominant at temperatures higher than 750 K, compared against the dominant status of triplet species. Therefore, the only singlet channel we considered here is directly branching to bimolecular CH_2CO+H_2 via R1-5.

The rate coefficients and corresponding branching ratios for the five reactions (from R1-1 to R1-5) in present model are plotted

in Fig. 5, compared against the results reported by Baulch et al. [60] and the data in AramcoMech2.0. As shown in Fig. 5(a), R1-1 and R1-5 show significant fall-off at temperatures higher than 800 K, which lead to great discrepancies as previous parameters are used. R1-1 and R1-2 are dominant pathways at low temperatures, and R1-3 is not a favored channel at temperatures lower than 1100 K. However, R1-3 becomes more important than R1-1 and is close to R1-2 at high temperatures due to the obvious fall-off region of R1-1. Therefore, an intersection between R1-1

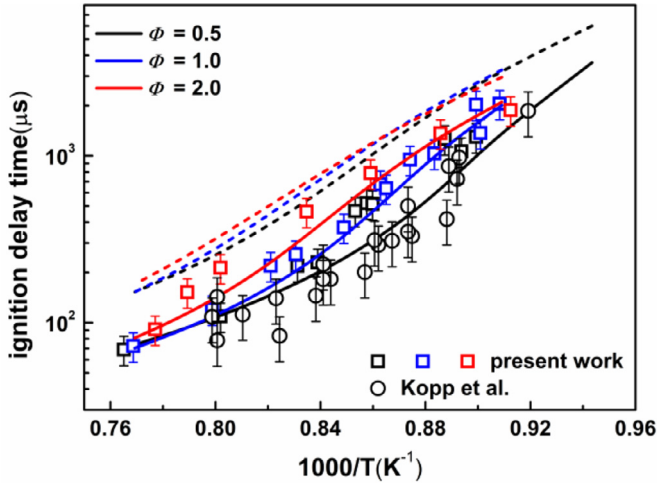


Fig. 6. Effect of the pathways and relevant rate coefficients of reaction class $C_2H_4 + \dot{O}$ on the ignition delay time for ethylene/air mixtures at high temperatures, a pressure of 1.0 atm, and $\phi = 0.5, 1.0$ and 2.0 , respectively. Symbols: experimental data in the present work and Kopp et al. [20] Solid lines: developed model based on the *ab initio* calculation of Li et al. [62] Dashed lines: simulation using the pathways and rate constants recommended by Baulch et al. [60].

and R1–3 is observed in Fig. 5(a) and(b) within 1100 K–1150 K. Furthermore, the rate coefficient of R1–5 from the previous recommendation is about an order of magnitude higher than the value of Li et al. [62] at 1500 K and the difference decreases as temperature decreasing. As Fig. 5(b) shows, the branching ratios of $C_2H_4 + \dot{O}$ from Baulch et al. [60] and AramcoMech2.0 remain unchanged over the whole temperature range, but the branching ratios taken from Li et al. [62] is highly temperature-dependent. Specifically, the branching ratio of Baulch et al. [60] is 0.35/0.60/0.05 for R1–2/R1–1/R1–5, and AramcoMech2.0 revised the branching ratio to 0.45/0.55 for R1–2/R1–1 and neglected the pathway of R1–5. As for the present model, although the branching ratios of R1–4 and R1–5 are always lower than 0.05 indicating the importance is limited at combustion relevant conditions, the branching ratio changes greatly along the whole temperature range for other three reactions. At 500 K, the branching ratio is 0.26/0.66/0.03 for R1–2/R1–1/R1–3, whereas it changes to 0.45/0.13/0.40 for R1–2/R1–1/R1–3 at 1500 K, indicating the greatly increasing importance of R1–3 and decreasing importance of R1–1 as temperature changes from 500 K to 1500 K. As a result, two intersections of branching ratios arise from Fig. 5(b). The one is for R1–1 and R1–2 within 850 K – 900 K, the other is for R1–1 and R1–3 between 1100 K and 1150 K, which agrees well with the trend observed in Fig. 5(a). By replacing the five pathways in the developed model with the recommended three pathways (R1–1, R1–2 and R1–5) proposed by Baulch et al. [60], the effects of this reaction class on the ignition delay time in high temperatures are shown in Fig. 6. The misunderstanding regarding the rate coefficients and branching ratios as discussed above leads to a remarkable over-prediction of the ignition delay times at high temperatures. However, the developed model captures the experimental trend well at $\phi = 0.5$ – 2.0 due to the adoption of the reasonable kinetic parameters obtained by AITSTM (ab initio transition state theory based master equation) approach from Li et al. [62].

3.2. $C_2H_5 + O_2$, $C_2H_4 + HO_2$ and $\dot{C}_2H_4OOH + O_2$

- $C_2H_4 + \dot{H} = \dot{C}_2H_5$ (R2–1)
- $\dot{C}_2H_5 + O_2 = C_2H_4 + HO_2$ (R2–2)
- $\dot{C}_2H_5 + O_2 = C_2H_5OO\dot{O}$ (R2–3)
- $C_2H_5OO\dot{O} = C_2H_4 + HO_2$ (R2–4)

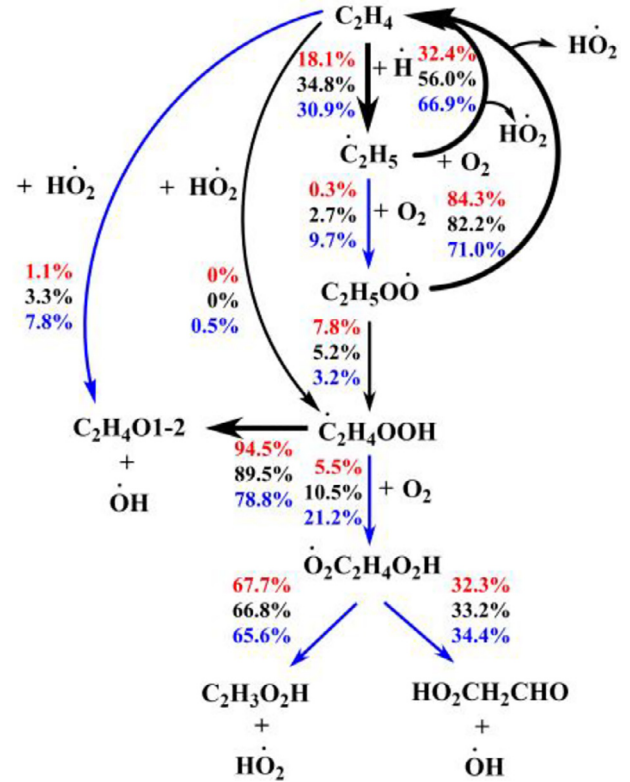


Fig. 7. Flux analysis of $\dot{C}_2H_5 + O_2$, $C_2H_4 + HO_2$ and $\dot{C}_2H_4OOH + O_2$, for ethylene/air mixtures at $p = 10$ atm, $\phi = 1.0$ and 20% fuel consumption. Red numbers: 1200 K; black numbers: 950 K; blue numbers: 750 K. The most dominant channels are highlighted by bold arrows, and the channels favored only for low-temperature combustion are marked by blue arrows. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

- $C_2H_5OO\dot{O} = \dot{C}_2H_4OOH$ (R2–5)
- $C_2H_4 + HO_2 = \dot{C}_2H_4OOH$ (R2–6)
- $C_2H_4 + HO_2 = C_2H_4O1-2 + \dot{OH}$ (R2–7)
- $C_2H_4 + HO_2 = \dot{C}_2H_3 + H_2O_2$ (R2–8)
- $\dot{C}_2H_4OOH = C_2H_4O1-2 + \dot{OH}$ (R2–9)
- $\dot{C}_2H_4OOH + O_2 = \dot{O}_2C_2H_4O_2H$ (R2–10)
- $\dot{O}_2C_2H_4O_2H = C_2H_3O_2H + HO_2$ (R2–11)
- $\dot{O}_2C_2H_4O_2H = HO_2CH_2CHO + \dot{OH}$ (R2–12)

For the high concentration of hydroperoxyl radicals (HO_2) at the pre-ignition regime, reactions between C_2H_4 and HO_2 are influential at elevated pressures and low-to-intermediate temperatures [63]. $C_2H_4 + HO_2$ is always in connection with $\dot{C}_2H_5 + O_2$ located in $C_2H_5O_2$ potential energy surface (PES), as $\dot{C}_2H_5$ is easily formed by addition reaction between C_2H_4 and $\dot{H}$ (R2–1). After the formation of hydroperoxyethyl radical ($\dot{C}_2H_4OOH$), the subsequent oxidation can occur via R2–10. Thus, it is necessary to discuss the comprehensive effect of reaction classes including $C_2H_4 + HO_2$, $\dot{C}_2H_5 + O_2$ and $\dot{C}_2H_4OOH + O_2$, and the temperature-dependent flux analysis is carried out employing the developed model to determine major consumption pathways, as shown in Fig. 7.

For the addition reactions of hydroperoxyl radicals, the well-skipping reaction to form oxirane (C_2H_4O1-2) and hydroxyl radical via reaction R2–7 is the dominant pathway, especially in low temperatures, which accounts for 7.8% of the total flux on ethylene consumption. By contrast, the direct combination reaction (R2–6) is actually not favored over the whole temperature range, contributing almost 0% of ethylene consumption as shown in Fig. 7. Therefore, the most dominant mechanism forming oxirane is the direct reaction through R2–7, instead of through

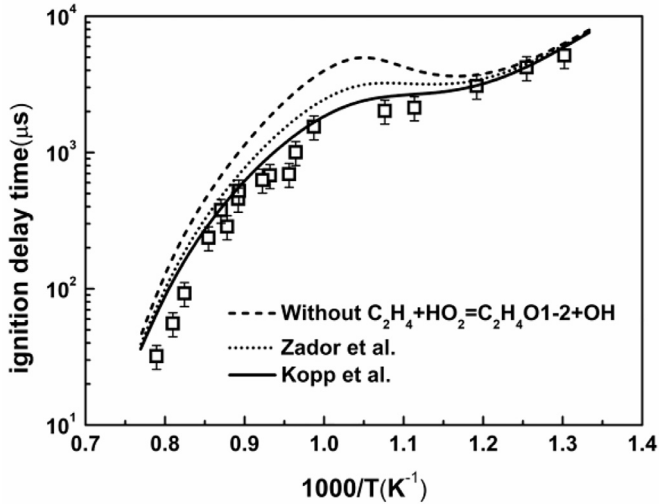


Fig. 8. Effect of well-skipping reaction $\text{C}_2\text{H}_4 + \text{H}_2\text{O}_2 = \text{C}_2\text{H}_4\text{O}_1-2 + \text{OH}$ on the ignition delay time for ethylene/air mixtures at $p = 10$ atm, $\phi = 1.0$. The rate coefficients are taken into the developed model from Zádor et al. [63] and Kopp et al. [35], respectively. Symbols: experimental data; Lines: simulations.

R2-6 and R2-9. Reaction R2-7 has a great promoting effect on intermediate-temperature oxidation, which has been determined in sensitivity analyses as shown in Figs. 14 and 15. Figure 8 illustrates that the absence of this well-skipping reaction will produce an abnormal negative-temperature-coefficient (NTC) region for ethylene oxidation process at a typical pressure of 10 atm and $\phi = 1.0$. Figure 8 also presents the modeling results by using the rate constant for R2-7 from Zádor et al. [63] and Kopp et al. [35], respectively. The simulation result employing the theoretical work of Zádor et al. [63] may over-predict the ignition delay time at intermediate-temperature region. However, rate coefficient taken from the recommendation of Kopp et al. [35] can better meet the experimental results, which are finally decided to be adopted in the present model.

Instead of direct combination of R2-6, $\dot{\text{C}}_2\text{H}_4\text{OOH}$ is actually formed by the isomerization of ethylperoxy radical ($\text{C}_2\text{H}_5\text{OO}\cdot$) through R2-5, following the first O_2 addition to $\dot{\text{C}}_2\text{H}_5$ through R2-3, which account for 9.7% of $\dot{\text{C}}_2\text{H}_5$ consumption at 750 K, but it is not dominant at 950 K and 1200 K. It should be noted that the most dominant consumption channel of $\dot{\text{C}}_2\text{H}_5$ is to react back to C_2H_4 plus HO_2 through the well-skipping reaction R2-2, especially in 750 K which account for 66.9% of the total flux to $\dot{\text{C}}_2\text{H}_5$ consumption. Therefore, this reaction shows strong inhibiting effect for fuel reactivity because of the formation of C_2H_4 and stable HO_2 radical. The similar impact can be found for dissociating reaction R2-4, which accounts for 84.3%, 82.2% and 71.0% of $\text{C}_2\text{H}_5\text{OO}\cdot$ consumption in 1200 K, 950 K and 750 K, respectively. Recently, three research groups have investigated $\text{C}_2\text{H}_4 + \text{H}_2\text{O}_2$ and $\text{C}_2\text{H}_5 + \text{O}_2$ PES including DeSain et al. [64], Zádor et al. [63], and Guo et al. [65]. Except for R2-7 discussed above, the reactions on the whole PES are added into the developed model to determine their comprehensive effect on ignition delay times at a typical pressure of 10 atm and $\phi = 1.0$. As shown in Fig. 9, the modeling results by using the rate coefficients from Zádor et al. [63] and Guo et al. [65] slightly over-predict the ignition delay times in high and intermediate temperatures, and the differences between them at low temperatures can be ignored. By contrast, the kinetic parameters from the theoretical work of DeSain et al. [64], in accordance with AramcoMech2.0 [30], can better capture the experimental data.

At 1200 K, 94.5% of total flux on $\dot{\text{C}}_2\text{H}_4\text{OOH}$ consumption is to form $\text{C}_2\text{H}_4\text{O}_1-2$ plus OH radical (R2-9). However, as the temperature decreases from 1200 K to 750 K, flux ratio of R2-9 decreases

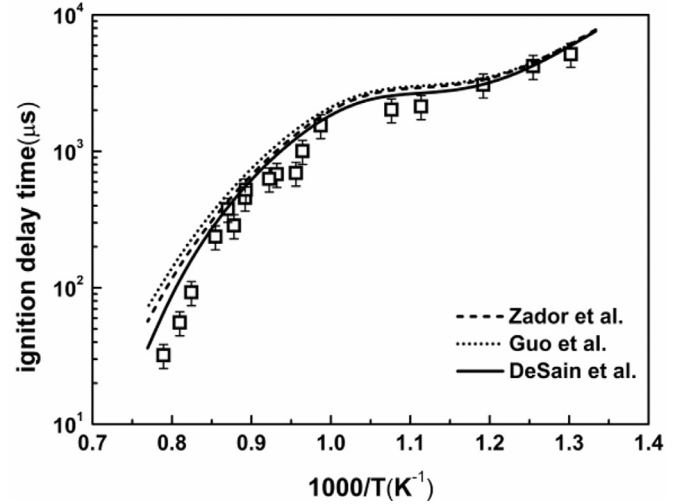


Fig. 9. The comprehensive effect of the whole $\text{C}_2\text{H}_5\text{O}_2$ PES on the ignition delay time for ethylene/air mixtures at $p = 10$ atm, $\phi = 1.0$, including well-to-well and well-skipping reactions except for R2-7 which has been discussed in Fig. 8. The rate coefficients are taken into the developed model from Zádor et al. [63], Guo et al. [65] and DeSain et al. [64], respectively. Symbols: experimental data; Lines: simulations.

to 78.8%, while that of second O_2 addition (R2-10) increases from 5.5% to 21.2%. Thus, the second O_2 addition reaction (R2-10) and subsequent dissociation reactions (R2-11 and R2-12) are considered in this work and added into the developed model, kinetic parameters are taken from the theoretical study by Bozzelli and Sheng [66]. Due to the further dissociation for vinyl hydroperoxide ($\text{C}_2\text{H}_3\text{O}_2\text{H}$) and hydroperoxide acetaldehyde ($\text{HO}_2\text{CH}_2\text{CHO}$) to produce the second OH radicals, the overall reactions can be regarded as a chain-branching process. As shown in Fig. 7, R2-11 accounts for 67.7%, 66.8% and 65.6% of the total flux on $\dot{\text{O}}_2\text{C}_2\text{H}_4\text{O}_2\text{H}$ consumption at 1200 K, 950 K and 750 K respectively, while R2-12 account for 32.3%, 33.2% and 34.4% respectively at these three typical conditions.

3.3. β -hydroxyethyl radical chemistry

- $\text{C}_2\text{H}_4 + \text{OH} = \text{P}\dot{\text{C}}_2\text{H}_4\text{OH}$ (R3-1)
- $\text{P}\dot{\text{C}}_2\text{H}_4\text{OH} + \text{O}_2 = \dot{\text{O}}_2\text{C}_2\text{H}_4\text{OH}$ (R3-2)
- $\text{P}\dot{\text{C}}_2\text{H}_4\text{OH} + \text{O}_2 = \text{C}_2\text{H}_3\text{OH} + \text{H}\dot{\text{O}}_2$ (R3-3)
- $\text{P}\dot{\text{C}}_2\text{H}_4\text{OH} + \text{O}_2 = \text{HOCYCOC} + \dot{\text{O}}\text{H}$ (R3-4)
- $\text{P}\dot{\text{C}}_2\text{H}_4\text{OH} + \text{O}_2 = 2\text{CH}_2\text{O} + \dot{\text{O}}\text{H}$ (R3-5)
- $\dot{\text{O}}_2\text{C}_2\text{H}_4\text{OH} = \text{HO}_2\dot{\text{C}}_2\text{H}_3\text{OH}$ (R3-6)
- $\dot{\text{O}}_2\text{C}_2\text{H}_4\text{OH} = \text{HO}_2\text{C}_2\text{H}_4\dot{\text{O}}$ (R3-7)
- $\text{HO}_2\text{C}_2\text{H}_4\dot{\text{O}} = 2\text{CH}_2\text{O} + \dot{\text{O}}\text{H}$ (R3-8)
- $\text{HO}_2\dot{\text{C}}_2\text{H}_3\text{OH} = \text{HOCYCOC} + \dot{\text{O}}\text{H}$ (R3-9)
- $\text{HO}_2\dot{\text{C}}_2\text{H}_3\text{OH} = \text{C}_2\text{H}_3\text{OH} + \text{H}\dot{\text{O}}_2$ (R3-10)
- $\text{HOCYCOC} = \dot{\text{C}}\text{H}_2\text{CHO} + \dot{\text{O}}\text{H}$ (R3-11)

Sarathy et al. [12,13] have proposed detailed reaction classes for alcohol combustion chemistry, and it was emphasized that the addition reaction between hydroxyalkyl radical and oxygen molecule has a great impact on controlling the system reactivity in low temperatures, as it is the first step leading to low-temperature chain branching process. Recently, the important role played by alcohol reactions on low-temperature combustion for alkenes has been found by many studies [30,31,57,67], because various hydroxyalkyl radicals are formed by addition between fuel and hydroxyl radical with highly exothermic characteristic. This reaction class is extensively supplemented and updated with accurate kinetics parameters in present work. Figure 10 gives the flux analysis employing the developed model at typical conditions. Regarding the

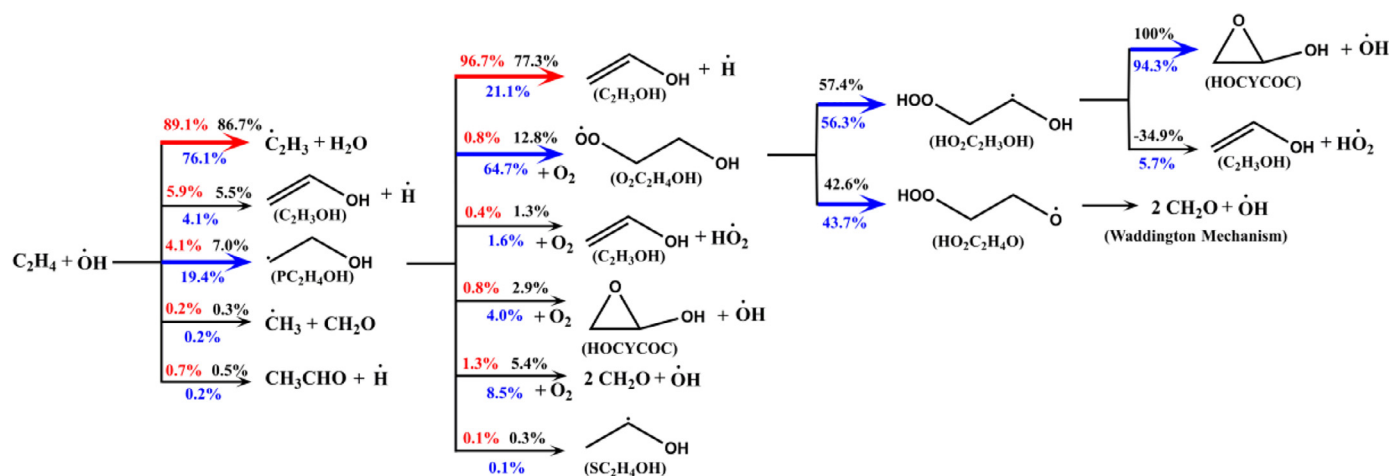


Fig. 10. Flux analysis for $\dot{O}H$ addition to ethylene and subsequent oxidation reactions using the developed model, for ethylene/air mixtures at $p = 10$ atm and $\phi = 1.0$ for 20% fuel consumption. Red numbers: 1200 K; black numbers: 950 K; blue numbers: 750 K. Bold blue arrows indicate the predominant channels for low-temperature combustion, and bold red arrows mark the channels for high-temperature. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

consumption of $C_2H_4 + \dot{O}H$, at low-to-high temperature range, hydrogen abstraction to form vinyl radical ($\dot{C}_2H_3$) and H_2O is the most dominant pathway. However, as the temperature changes, great flux differences appear for the formation of β -hydroxyethyl (PC_2H_4OH) via R3-1. This channel accounts for 19.4% on $C_2H_4 + \dot{O}H$ consumption at 750 K, compared to only 4.1% at 1200 K and 7.0% at 950 K, highlighting its high reactivity in ethylene low-temperature combustion. Another important channel forming $C_2H_3OH + \dot{H}$ accounts for 5.9%, 5.5% and 4.1% of the total flux on $C_2H_4 + \dot{O}H$ at 1200 K, 950 K and 750 K, respectively, its inhibiting effect is obvious by the sensitivity analysis in present work.

The subsequent oxidation steps after the formation of β -hydroxyethyl is O_2 addition to yield a chemical activated adduct of $[O_2C_2H_4OH]^*$, which can stabilize to form $\dot{O}_2C_2H_4OH$ radical via R3-2, or directly react to products via reactions R3-3 to R3-5. At high temperature of 1200 K, almost all of PC_2H_4OH forms ethenol (C_2H_3OH) by elimination of $\dot{H}$, which accounts for 96.7% of the total flux on PC_2H_4OH consumption. As temperature decreases to 750 K, the stabilization of $[O_2C_2H_4OH]^*$ becomes the most dominant channel contributing 64.7% of PC_2H_4OH consumption. Two isomerization pathways are considered for the consumption of $\dot{O}_2C_2H_4OH$. The first channel undergoes 1,5-H migration to form $HO_2CH_2CH_2\dot{O}$ via R3-7, followed by rapid β -scission to produce $\dot{C}H_2OOH$ and CH_2O . The unstable intermediate $\dot{C}H_2OOH$ continues to dissociate to CH_2O and $\dot{O}H$ also by rapid β -scission. This whole process is known as Waddington mechanism [68], and accounts for 42.6% and 43.7% of the total flux on $\dot{O}_2C_2H_4OH$ dissociations at 950 K and 750 K, respectively. Another channel undergoes 1,4-H migration from the hydroxyl carbon site to the peroxy oxygen with the formation of $HO_2\dot{C}_2H_3OH$ via R3-6, which consumes 57.4% and 56.3% of $\dot{O}_2C_2H_4OH$, at 950 K and 750 K, respectively. The Waddington mechanism for ethylene combustion is actually a greatly inhibiting reaction, and will be explained in Section 4.2. Two low-energy pathways are responsible for the consumption of $HO_2\dot{C}_2H_3OH$. One pathway involves the formation of hydroxyl oxirane ($HOCYCOC$) with the elimination of $\dot{O}H$ via reaction R3-9, and the other pathway involves the formation of ethenol (C_2H_3OH) by β -scission with the elimination of HO_2 via reaction R3-10. $HOCYCOC$ can further dissociate to $\dot{C}H_2CHO$ plus $\dot{O}H$ via ring-opening reaction R3-11, therefore can be considered as a chain-branching process promoting reactivity of reaction system. At an intermediate temperature of 950 K, almost all of $HO_2\dot{C}_2H_3OH$ dissociate to

form $HOCYCOC$ and $\dot{O}H$ radical. One interesting phenomenon is found for R3-10. At a low temperature of 750 K, it consumes 5.7% of $HO_2\dot{C}_2H_3OH$. However, at an intermediate temperature of 950 K, it contributes to 34.9% formation of $HO_2\dot{C}_2H_3OH$, which exhibits opposite effect compared against 750 K. For the low-reactivity of HO_2 , forward and reverse channels for R3-10 are both inhibiting reactions at a wide temperature range, as shown in Fig. 14.

Two studies computed the PES of β -hydroxyethyl + O_2 for ethanol combustion. Silva and Bozzelli [69] gave the complete PES for β -hydroxyethyl + O_2 and α -hydroxyethyl + O_2 using G3B3 composite method. Meanwhile pressure-dependent rate coefficients were obtained using QRRK/ME, which are the only available parameters from literature as we know. Zádor et al. [70] recently investigated the same PES employing both theoretical analysis and experimental product study. Unfortunately, the high-precision elementary rate coefficients and branching ratios for β -hydroxyethyl + O_2 were still uncertain due to the multi-reference character of certain transition states. In this work, kinetic parameters are taken from Silva et al. [69], and optimization is simultaneously carried out to improve the model predictability. From the work of Zádor et al. [70], in consideration of T_1 diagnostic values for all transition states, only that of Waddington mechanism (R3-7) surpassed the limiting value of 0.045, which reached 0.115 by QCISD(T)/cc-pV ∞ Z. The very high T_1 diagnostic value means great multi-reference effect of reaction R3-7 leading to serious discrepancy for energy calculation. The barrier heights for the Waddington mechanism using different single-reference and multi-reference methods in the literature are concluded in Table 3. The energy calculated by G3B3 is 4.3 kcal/mol lower than that of high-level Davidson corrected MRCI(3e,3o)/cc-pVQZ, while only 0.63 kcal/mol lower than CASPT2(3e,3o)/cc-pVQZ. QCISD(T)/cc-pV ∞ Z method shows the highest energy of 32.44 kcal/mol, which should be ignored for the great T_1 diagnostic value. The uncertainty for species with high T_1 diagnostics was estimated less than 5 kcal/mol in the work of Zádor et al. [70]. In order to meet the experimental data with simulation results, the activation energy of R3-7 ($\dot{O}_2C_2H_4OH = HO_2C_2H_4\dot{O}$) obtained by Silva et al. [69] is increased by 3 kcal/mol in the developed model to explain the multi-reference effect. Except for R3-7 with the greatest multi-reference effect, other reactions with slight multi-reference characteristic are not changed. The effect of revising kinetic parameter on ignition delay times over a wide temperature range is

Table 3

Barrier heights of R3–7 ($\dot{\text{O}}_2\text{C}_2\text{H}_4\text{OH}=\text{HO}_2\text{C}_2\text{H}_4\dot{\text{O}}$) obtained with different quantum methods [69,70]. Units in kcal/mol.

G3B3	QCISD(T)/cc-pV ∞ Z	CASPT2/cc-pVQZ	MRCI/cc-pVQZ	Davidson corrected MRCI/cc-pVQZ
25.8	32.44	26.43	32.53	30.11

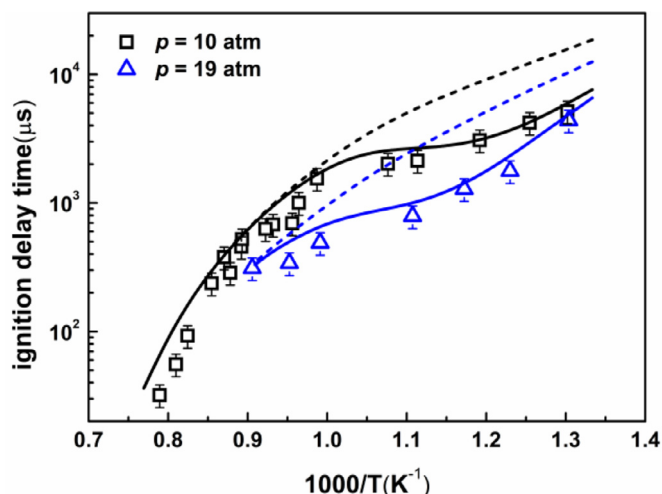


Fig. 11. Effect of the activation energy of Waddington-mechanism reaction ($\text{O}_2\text{C}_2\text{H}_4\text{OH}=\text{HO}_2\text{C}_2\text{H}_4\dot{\text{O}}$) on the ignition delay times for ethylene/air mixtures at $p = 10$ and 19 atm, $\phi = 1.0$. Symbols: experimental data. Dashed lines: simulation with original data from Silva and Bozzelli [69]. Solid lines: simulation with activation energy increased by 3 kcal/mol.

given in Fig. 11 at pressures of 10 atm and 19 atm, and at $\phi = 1.0$. Significant modeling improvement is achieved at low-intermediate temperatures (< 1000 K), while no obvious changes are observed at higher temperatures. Above optimization illustrates the importance of considering the multi-reference characteristic of reactions in mechanism construction, especially for β -hydroxyethyl oxidation. Further experimental and theoretical studies on this reaction class employing state-of-the-art methods will be helpful to determine the accurate kinetic parameters in future.

3.4. Other sub-mechanisms

Some pathways and the corresponding rate coefficients of important oxidation intermediates are updated or added into current model. Details for these reactions can be found in Table S2, and sequence number is marked for each one. Hydroperoxide acetaldehyde ($\text{HO}_2\text{CH}_2\text{CHO}$) is an intermediate species from the dissociation of $\dot{\text{O}}_2\text{C}_2\text{H}_4\text{O}_2\text{H}$. It consumes through two pathways, the one is direct dissociation of the O–OH bond, followed by rapid β -scission to form $\text{CH}_2\text{O}+\text{HCO}+\dot{\text{O}}\text{H}$ (No. 10). Another is H-abstraction reaction at different sites by small radicals X ($\text{X} = \text{O}_2, \dot{\text{H}}, \dot{\text{O}}\text{H}, \dot{\text{O}}, \text{H}\dot{\text{O}}_2, \text{HCO}, \dot{\text{C}}\text{H}_3, \dot{\text{C}}_2\text{H}_5$) (No. 59–61), whose rate coefficients are taken from estimation of Ranzi et al. [29] Vinyl hydroperoxide ($\text{C}_2\text{H}_3\text{O}_2\text{H}$) and ethenol ($\text{C}_2\text{H}_3\text{OH}$) are important oxygenated intermediates in ethylene oxidation, and the addition and dissociation reactions are both considered in the developed model. The sub-mechanism for $\text{C}_2\text{H}_3\text{O}_2\text{H}$ is obtained from the theoretical study of Guo et al. [71] (No. 27–29) The barrierless addition to $\dot{\text{O}}\text{H}$ by $\dot{\text{C}}_2\text{H}_3$ producing ethenol (No. 30) is added into the model based on the work of Knyazev [72]. The sub-mechanism on the important oxygenated products of oxirane and oxiranyl radical (No. 43–58) is taken from the work of Joshi et al. [73], and that of the reactions on formaldehyde (No. 31–42) is obtained by Christensen et al. [74] Further

more, since α -hydroxyethyl can be formed by isomerization of β -hydroxyethyl, its oxidation PES is also added into model [69].

4. Model validation and analyses

4.1. Ignition delay time validations

The developed model is comprehensively validated against experimental results of ignition delay times under a wide range of conditions, as shown in Table 4. Validations involve ethylene/air mixtures in this work and Ar-dilution conditions from recent shock tube study of Saxena et al. [17].

Figure 12 shows the high-temperature simulation results for ethylene/air mixtures at pressures of 1 atm– 10 atm, and equivalence ratios of $0.5, 1.0$ and 2.0 . At stoichiometric and fuel-rich conditions, as shown in Fig. 12b and c, the ignition delay time decreases as the pressure increases, indicating high reactivity at higher pressure. This phenomenon can be explained by the increased reactant concentrations as the pressure increases. However, at fuel-lean condition, the pressure effect becomes weak, and two turning points of ‘cross-over’ arise. This phenomenon of ‘cross-over’ was also found by the similar study for ethylene ignition [20], and for other higher hydrocarbons [75,76], which can be attributed to kinetic coupling resulting from the changes in radical pool concentration [77]. The developed model is able to capture the experimental data well, including both traditional pressure effect and ‘cross-over’ phenomenon at fuel-lean condition. However, slight under-predictions are also observed at $\phi = 0.5$, and more investigations are needed for process of fuel-lean oxidation. Comparisons of simulation results between the developed model and contemporary models from literature are given in Figs. S4 and S5 of Supplementary material. Effects of pressure and dilution ratios for Ar-dilution conditions in high temperature range are also given in Figs. S6 and S7. Similar conclusion can be drawn, but no ‘cross-over’ phenomenon is observed for Ar-dilution conditions.

Model prediction for ethylene/air mixtures at high pressures over a wide range of temperatures is presented in Fig. 13 with the effect of equivalence ratio. There is no definite NTC, but the ignition data at low temperatures are extremely lower than the extrapolation from high temperatures, which agrees well with the phenomenon reported by Cadman et al. [21] Compared with Fig. 3, present model can predict the ignition behavior at low-intermediate temperatures (720 – 1000 K). Effect of equivalence ratio is obvious in NTC region, where ignition delay times decrease with equivalence ratio increasing. This can be explained by the dominant role played by β -hydroxyethyl generated from the addition of hydroxyl to the double bond of ethylene ($\text{C}_2\text{H}_4+\dot{\text{O}}\text{H}=\text{PC}_2\text{H}_4\text{OH}$). As concentration of fuel increases, the effect of low-temperature chemistry based on β -hydroxyethyl temperature becomes stronger. However, at higher temperatures, the difference under various equivalence ratios is much smaller, and ‘cross-over’ occurs at temperature over 1250 K. This is due to the increasing importance of C_0 chemistry, such as chain-branching reaction $\text{O}_2+\text{H}=\dot{\text{O}}+\text{OH}$, which would take place of reactions related to C_2 chemistry at very high temperatures. This behavior is well predicted by the developed model.

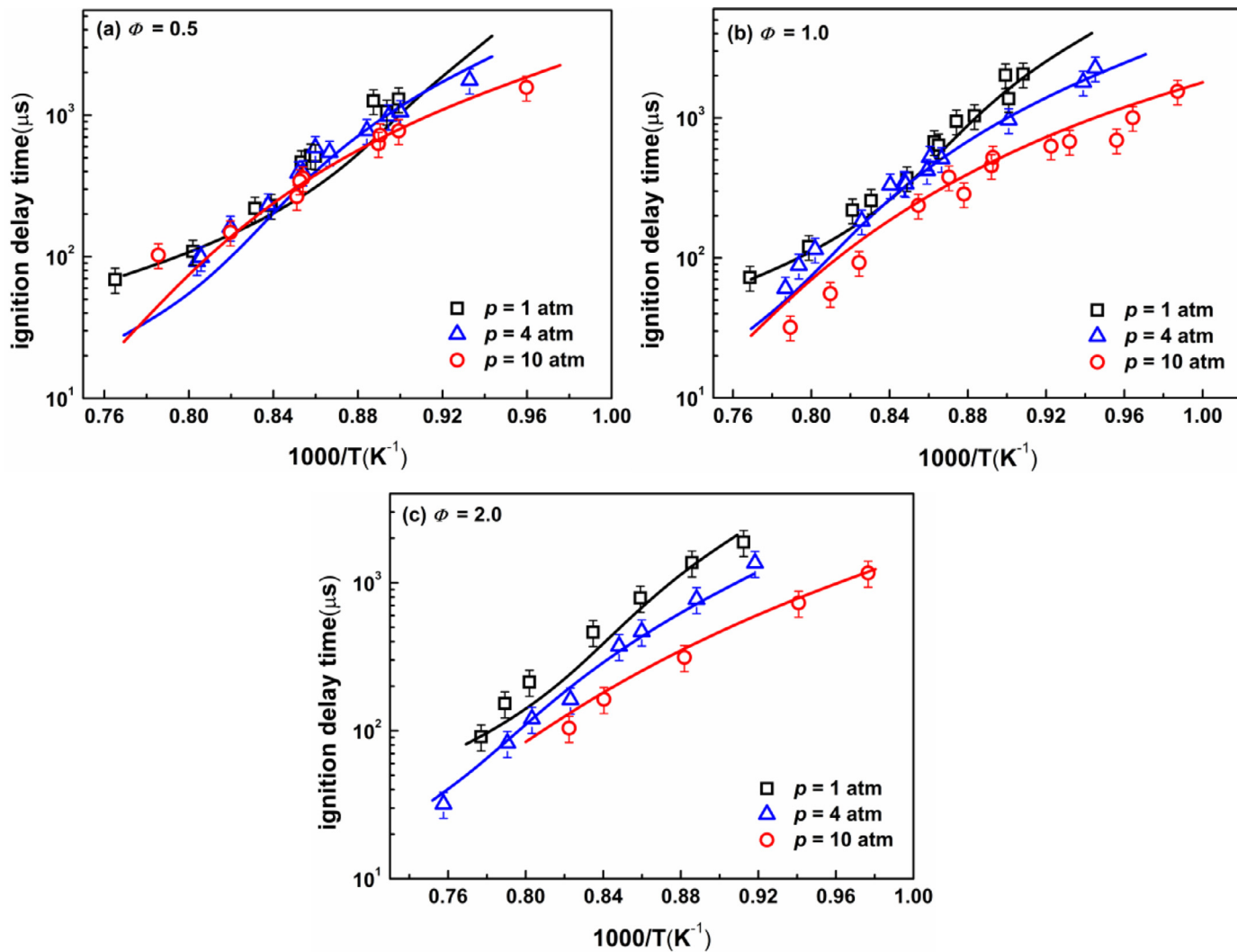


Fig. 12. Influence of pressure on ignition delay time for ethylene/air mixtures at pressures of 1, 4 and 10 atm. (a) $\phi = 0.5$; (b) $\phi = 1.0$; (c) $\phi = 2.0$. Symbols: experimental data in present work. Lines: simulations employing the developed model.

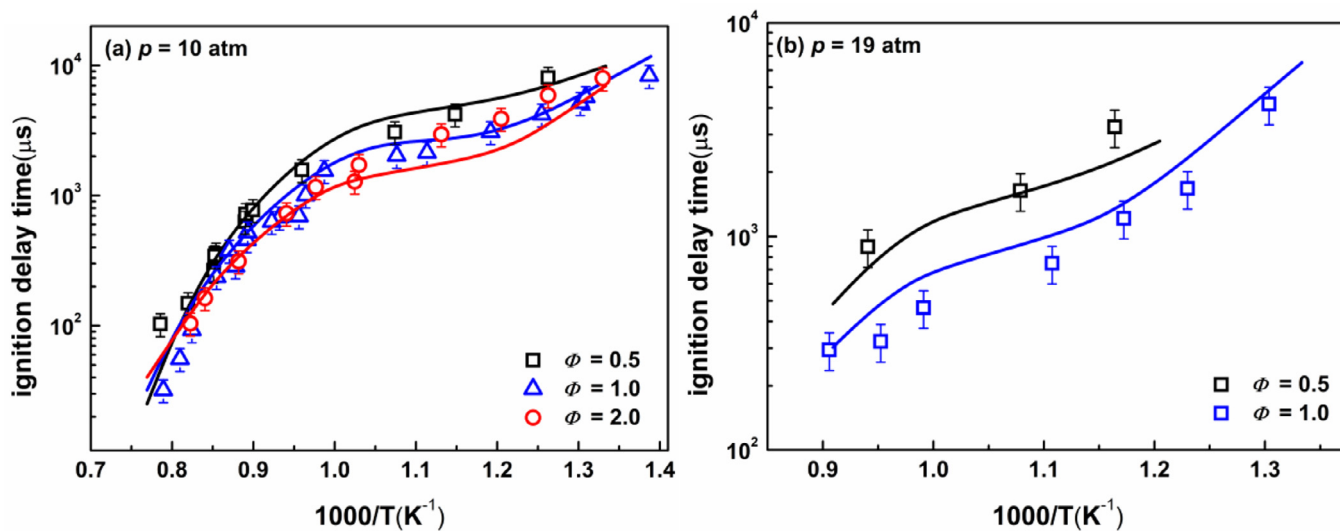


Fig. 13. Influence of equivalence ratio on ignition delay time for ethylene/air mixtures over a wide range of temperatures. (a) $p = 10$ atm; (b) $p = 19$ atm. Symbols: experimental data in present work. Lines: simulations employing the developed model.

Table 4
Validation conditions of ignition delay times in present study.

Pressure (atm)	Equivalence ratio	Dilution	Temperature (K)	Refs.
1, 4	0.5, 1, 2	Air	1058 - 1320	This work
10	0.5, 1, 2	Air	721 - 1273	This work
19	0.5, 1	Air	767 - 1104	This work
2, 10	3	93% Ar ^a	1064 - 1579	[17]
2, 10	1	96% Ar	1197 - 1690	[17]
10	1	98% Ar	1223 - 1719	[17]

^a The mole fraction of Ar in ethylene/O₂/Ar mixtures.

4.2. Sensitivity analysis

Brute-force sensitivity analysis is carried out to determine the key reactions responsible for fuel consumption under certain conditions by increasing the rate coefficients of every reaction by a factor of two resulting in the effect on predicted ignition delay times. Specifically, the sensitivity coefficient is defined as follows:

$$S = \frac{\tau(2k_i) - \tau(k_i)}{\tau(k_i)}$$

This equation form is firstly presented by Kumar et al. [22], where τ is the ignition delay time, k_i is the rate coefficient of the i th reaction, and $2k_i$ indicates increasing k_i by a factor of two. A positive value of S suggests that the corresponding reaction inhibits the reactivity of the ignition process, and vice versa.

The key reactions for the ignition delay time for ethylene/air mixtures are highlighted by sensitivity analyses as displayed in Figs. 14 and 15 under different conditions. Several top reactions with large positive and negative sensitivity coefficient are plotted here, and the coefficients of all reactions can be found in Supplementary material. At a low temperature of 750 K, the most important reactions are $\dot{\text{O}}_2\text{C}_2\text{H}_4\text{OH}=\text{HO}_2\text{C}_2\text{H}_4\dot{\text{O}}$ (R3-7) and $\dot{\text{O}}_2\text{C}_2\text{H}_4\text{OH}=\text{HO}_2\dot{\text{C}}_2\text{H}_3\text{OH}$ (R3-6) as shown in Fig. 14(a), which inhibit and promote the reactivity, respectively. The competition between these two pathways for β -hydroxyethyl consumption leads to a change in the slope of the ignition delay curve at the region of low temperatures, as given in Fig. 13. After the formation of $\text{HO}_2\text{C}_2\text{H}_4\dot{\text{O}}$, it can rapidly dissociate to two CH_2O plus one $\dot{\text{O}}\text{H}$ (R3-8). Although active radical $\dot{\text{O}}\text{H}$ is formed, it is immediately consumed by H-abstraction reaction with CH_2O to produce HCO radical plus H_2O ($\text{CH}_2\text{O}+\dot{\text{O}}\text{H}=\text{HCO}+\text{H}_2\text{O}$). Regarding the consumption of the reactive hydroxyl radical and the formation of the stable oxidation product of water, the H-abstraction $\text{CH}_2\text{O}+\dot{\text{O}}\text{H}=\text{HCO}+\text{H}_2\text{O}$ is an important inhibiting reaction at 750 K and 950 K, as shown in Fig. 14(a) and (b). This phenomenon can explain the strong inhibiting role played by the Waddington mechanism of $\dot{\text{O}}_2\text{C}_2\text{H}_4\text{OH}=\text{HO}_2\text{C}_2\text{H}_4\dot{\text{O}}$ (R3-7) followed by the formation of CH_2O and $\dot{\text{O}}\text{H}$. The promoting effect of $\dot{\text{O}}_2\text{C}_2\text{H}_4\text{OH}=\text{HO}_2\dot{\text{C}}_2\text{H}_3\text{OH}$ (R3-6) is caused by the further dissociation through the reaction of $\text{HO}_2\dot{\text{C}}_2\text{H}_3\text{OH}=\text{HOCYCOC}+\dot{\text{O}}\text{H}$ (R3-9). The overall reaction can be considered as a chain-branching process due to the formation of the second $\dot{\text{O}}\text{H}$ after the ring-opening of HOCYCOC (R3-11). However, another consumption channel of $\text{HO}_2\dot{\text{C}}_2\text{H}_3\text{OH}$ via reaction $\text{HO}_2\dot{\text{C}}_2\text{H}_3\text{OH}=\text{C}_2\text{H}_3\text{OH}+\text{HO}_2\dot{\text{O}}$ (R3-10) exerts a significant inhibiting impact on reactivity of system due to the low-reactivity of $\text{HO}_2\dot{\text{O}}$ as discussed in Section 3.3. The results of sensitivity analyses illustrate that β -hydroxyethyl chemistry controls the low-temperature oxidation of ethylene. Similar results are observed at an intermediate temperature of 950 K, as shown in Fig. 14(b).

As temperature increases to 950 K and 1200 K, another reaction class of the addition between O_2 and vinyl radical ($\dot{\text{C}}_2\text{H}_3$) becomes extremely important. As shown in Fig. 14(b) and (c), the chain-branching reaction forming $\dot{\text{C}}_2\text{HCHO}+\dot{\text{O}}$ and chain-propagating reaction forming $\text{CH}_2\text{O}+\text{HCO}$ are dominant channels. It should be noted that stabilization of adduct vinylperoxy ($\dot{\text{C}}_2\text{H}_3+\text{O}_2=\text{C}_2\text{H}_3\text{O}_2$)

inhibits the reactivity, which has the opposite effect compared with stabilization of $\dot{\text{O}}_2\text{C}_2\text{H}_4\text{OH}$ ($\text{P}\dot{\text{C}}_2\text{H}_4\text{OH}+\text{O}_2=\dot{\text{O}}_2\text{C}_2\text{H}_4\text{OH}$). Reaction $\text{C}_2\text{H}_4+\text{HO}_2=\text{C}_2\text{H}_4\text{O1-2}+\dot{\text{O}}\text{H}$ (R2-7) is an important promoting reaction due to the formation of reactive oxirane and hydroxyl radical over a wide temperature range. Its effect on ignition delay times has been illustrated in Fig. 8.

At 1200 K, chain branching reaction $\text{O}_2+\dot{\text{H}}=\dot{\text{O}}+\dot{\text{O}}\text{H}$ becomes the most crucial promoting reaction, while $\dot{\text{C}}_2\text{H}_5+\text{O}_2=\text{C}_2\text{H}_4+\text{HO}_2$ greatly retards the oxidation process due to the formation of inactive ethylene and HO_2 . As Fig. 14(c) shows, the effect of pressure on dominant reactions is great at high temperatures. One typical example is the addition reactions between C_2H_4 and $\dot{\text{O}}$ atom, whose significant influence on high-temperature oxidation of ethylene has been discussed in Section 3.1. At a pressure of 1 atm, the influence of two competitive channels including $\text{C}_2\text{H}_4+\dot{\text{O}}=\dot{\text{C}}_2\text{H}_3+\text{HCO}$ and $\text{C}_2\text{H}_4+\dot{\text{O}}=\dot{\text{C}}_2\text{HCHO}+\dot{\text{H}}$ is obvious, compared against the weak effect at higher pressure of 10 atm. Other reactions, such as $\dot{\text{C}}_2\text{HCHO}(\text{+M})=\text{CH}_2\text{CO}+\dot{\text{H}}(\text{+M})$, $\text{C}_2\text{H}_4+\text{HO}_2=\dot{\text{C}}_2\text{H}_3+\text{H}_2\text{O}_2$, $\text{H}_2\text{O}_2(\text{+M})=2\dot{\text{O}}\text{H}(\text{+M})$ and $\text{HCO}+\text{M}=\dot{\text{H}}+\text{CO}+\text{M}$, are also sensitive to the change of pressure. Furthermore, Fig. 15 presents the effect of equivalence ratio at a typical pressure and temperature. However, the influence of equivalence ratio is weak for the dominant reactions.

4.3. Speciation validations of jet-stirred reactor

Jallais et al. [39] and Cong et al. [38] have measured the concentration profiles of stable species on ethylene oxidation process in jet-stirred reactor (JSR) under a wide range of temperatures from 770 K to 1300 K, and equivalence ratios from 0.5 – 10. The detailed validation conditions are given in Table 5. Figures 16 and 17 show the comparisons between experimental data and predictions by employing the developed model and different core-mechanisms from literature. Overall there is good agreement between the current model and experimental trend over a wide range of temperatures and equivalence ratios, especially for the concentration history of ethylene. As Fig. 16 shows, ethylene conversion starts around 780 K at equivalence ratios ranging from 3 to 10. Even though slight under-predictions are observed at temperatures lower than 840 K at $\phi = 3.0$, the developed model is able to capture the onset of fuel consumption. However, the temperature point where 100% of ethylene are consumed (i.e., the mole fraction of ethylene drops to zero) depends on equivalence ratio. As Fig. 17 shows, at fuel-lean condition of $\phi = 0.5$, the concentration of ethylene drops to zero after 1080 K. At $\phi = 1.0$, the totally-consumed point changes to higher than 1175 K, while ethylene continues to consume at 1300 K, $\phi = 2.0$. This is due to the decreasing reactivity as equivalence ratio rising. This trend is also well predicted by the developed model. The maximum points of concentration profiles for some intermediates are observed, such as CO , CH_4 , C_2H_2 and $\text{C}_2\text{H}_4\text{O1-2}$ (oxirane), as shown in Figs. 16 and 17. The turning point of CH_4 is around 1025 K, and the peak of concentration is 100 ppm at $\phi = 0.5$. As equivalence ratio increases to 2.0, the turning point changes to around 1150 K, and the peak

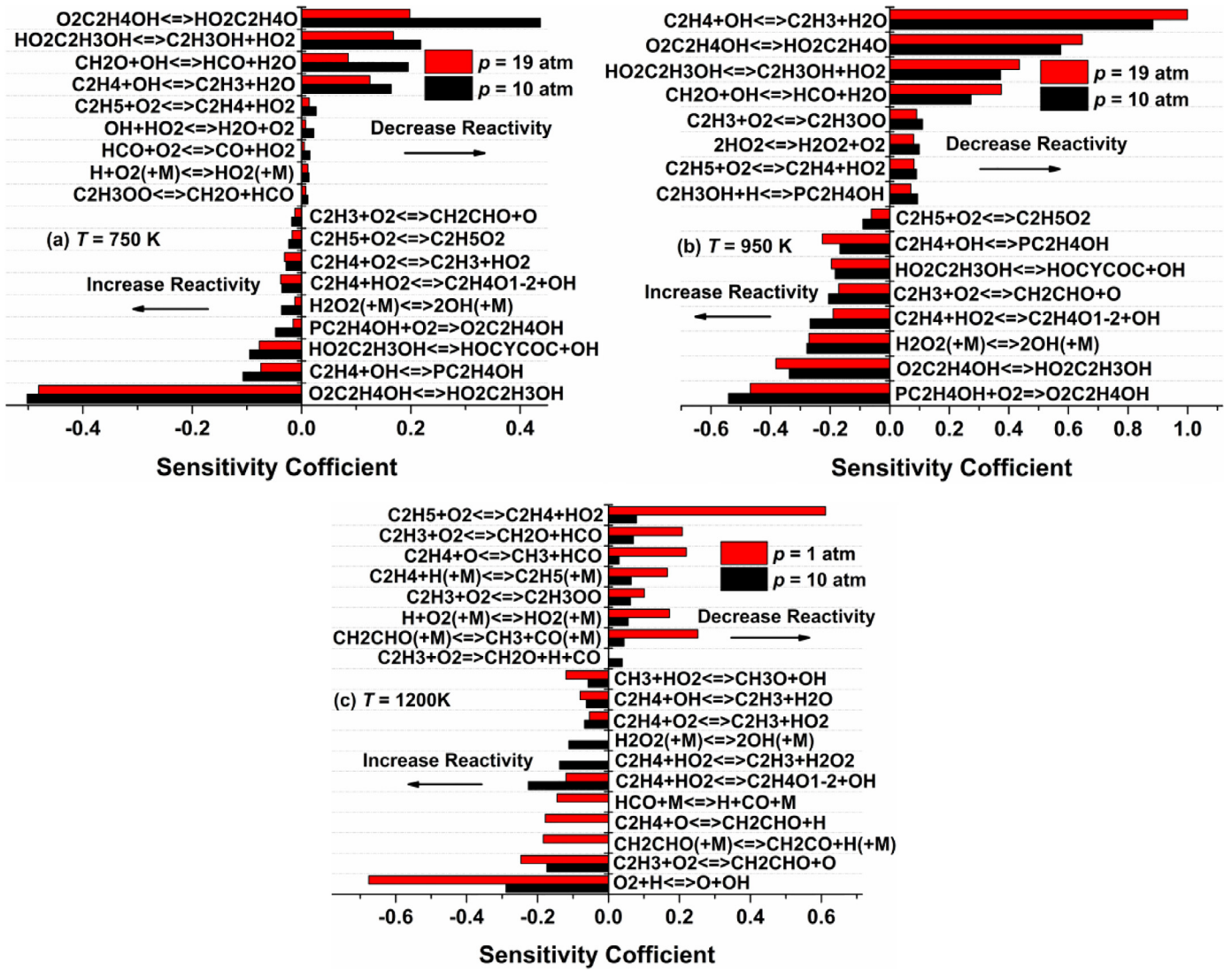


Fig. 14. Sensitivity analyses for ethylene/air mixtures at $\phi = 1.0$, and at temperatures of (a) 750 K; (b) 950 K and (c) 1200 K.

Table 5

Validation conditions of concentration profiles of stable species in JSR.

Reactant (Mole Fraction)	T (K)	p (atm)	ϕ	τ (s)	Ref.
0.05 C_2H_4 /0.05 O_2 /0.9 N_2	773–900	1	3	1.3	[39]
0.05 C_2H_4 /0.03 O_2 /0.92 N_2	773–900	1	5	1.3	[39]
0.05 C_2H_4 /0.015 O_2 /0.935 N_2	773–900	1	10	1.3	[39]
0.0017 C_2H_4 /0.0102 O_2 /0.9881 N_2	950–1100	1	0.5	0.12	[38]
0.0017 C_2H_4 /0.0051 O_2 /0.9932 N_2	950–1200	1	1	0.12	[38]
0.0017 C_2H_4 /0.00255 O_2 /0.99575 N_2	950–1300	1	2	0.12	[38]

of concentration increases to 150 ppm. The similar observation is made for C_2H_2 . As for CO and C_2H_4O1-2 , the changes of turning points are also obvious, but the maximum concentrations are almost constant at different conditions. Some under-predictions are found for concentration profiles of CO, CH_4 and C_2H_2 , and over-predictions are found for CO_2 at high temperatures. Unfortunately, none of existing models can accurately capture these data, thus further improvement is still needed for model development.

4.4. Laminar flame speed validation

Many researches [22,40–45] were carried out to determine the laminar flame speed for ethylene oxidation under various

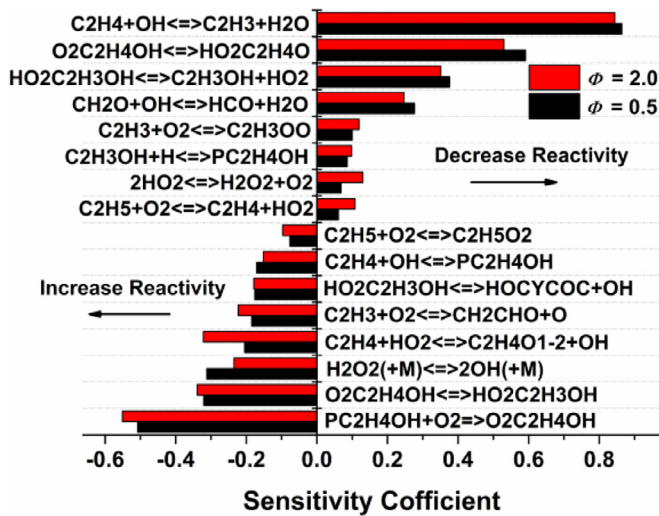
conditions. Table 6 gives all validation conditions and chosen experimental data from different sources.

Figure 18 shows the predicted laminar flame speed for ethylene/air mixtures using the present model and other popular core-mechanisms from literature [26,27,30], compared against the measurements at various pressures and temperatures. Figure 19 presents the effect of dilution ratios at 298 K and 1 atm. In general, the peak of laminar flame speed is predicted at the region from $\phi = 1.1$ to $\phi = 1.2$, which is independent of changing pressure, temperature and dilution ratio. The laminar flame speed decreases with pressure increasing, and increases with temperature increasing. Moreover, the laminar flame speed reduces as dilution ratio decreasing. This conclusion is easily acceptable because lower dilution ratio means the smaller partial

Table 6

Validation conditions and data sources of laminar flame speed for the developed model.

Authors	T (K)	p (atm)	Dilution ratio ^a	Equivalence ratio	Experimental method	Refs
Kumar et al.	298, 360, 470	1	21%	0.5 – 1.4	Counter-flow flame/DPIV ^b /linear extrapolation	[22]
Hirasawa et al.	298	1	21%	0.8 – 1.7	Counter-flow flame/ DPIV ^b /non-linear extrapolation	[42]
Jomaas et al.	298	1, 2, 5	21%	0.6 – 1.3	Outwardly propagating spherical flames/LDV ^c /non-linear extrapolation	[43]
Park et al.	298	1	21%	0.6 – 1.7	Counter-flow flame/ DPIV ^b /non-linear extrapolation	[44]
Egolfopoulos et al.	298	1	21%	0.5 – 2.2	Counter-flow flame/ DPIV ^b /linear extrapolation	[45]
Hassan et al.	298	2	21%	0.6 – 1.6	Outwardly propagating spherical flames/motion picture shadowgraphs/ non-linear extrapolation	[41]
Konnov et al.	298	1	14% – 18%	0.6 – 1.8	Non-stretched flames/heat flux method	[40]

^a $D=O_2/(O_2+N_2) \times 100\%$.^b DPIV: digital particle image velocimetry.^c LDV: laser doppler velocimetry.**Fig. 15.** The effect of equivalence ratios of sensitivity analyses for ethylene/air mixtures at $\phi = 0.5$ and 2.0 , at $p = 10$ atm, and at an intermediate temperature of 950 K.

pressure of reactant at a fixed pressure causing decreased reaction rate. As shown in Fig. 18(a), the experimental measurements taken from different sources have some differences, which can be attributed to different extrapolation methods addressing stretch effects [36], as shown in Table 6. Overall, different models have similar prediction results of laminar flame speed for ethylene/air combustion. Except for slight under-predictions observed at a high temperature of 470 K, the developed model is able to meet the experimental data under various conditions.

Figure 20 shows the sensitivity analyses of the flame speed under typical fuel-lean and fuel-rich conditions. Many important reactions shown here are from C_0 – C_1 mechanism. To some extent, satisfactory modeling results of laminar flame speed should be attributed to the detailed reaction network and accurate kinetic parameters of C_0 – C_1 core mechanism. At $\phi = 0.8$, the competition between chain branching reaction $O_2+H=\dot{O}+\dot{O}H$ and chain propagating reaction $\dot{H}+O_2(+M)=\dot{H}O_2(+M)$ controls the reactivity of laminar flame propagation. When the equivalence ratio increases to 1.2, the most important inhibiting reaction becomes the chain-termination reaction of $\dot{C}_2H_3+\dot{H}=C_2H_2+H_2$ due to the increasing concentration of ethylene. Some of C_2

chemistry discussed in Section 4.2 also have an important influence on laminar flame speed, such as vinyl radical oxidation ($\dot{C}_2H_3+O_2=\dot{C}_2H_2CHO+\dot{O}$), two promoting addition reactions between oxygen atom and ethylene ($C_2H_4+\dot{O}=\dot{C}_2H_2CHO+\dot{H}$ and $C_2H_4+\dot{O}=\dot{C}_2H_2(X^3B_1)+CH_2O$), and so on. A novel reaction about the singlet carbene of $CH_2(S)+O_2=\dot{H}+\dot{O}H+CO$ is observed at fuel-rich condition, which was not presented in previous studies. One notable phenomenon should be emphasized, i.e., molecules larger than two carbon atoms and low-temperature reaction class involved in the formation and consumption of peroxide do not show an obvious influence on the laminar burning velocity.

4.5. Effect of model revisions on ethanol and ethane oxidation

Above elaborations illustrate that ethylene oxidation is always in connection with ethanol (C_2H_5OH) oxidation through hydrox-ethyl chemistry, as well as ethane (C_2H_6) whose typical ethyl radical ($\dot{C}_2H_5$) can be easily formed from ethylene through C_2 chemistry, such as C_2H_4+H and $C_2H_4+HO_2$. Thus, the additions and updates of the developed model in this work will also affect the predictions of C_2H_5OH and C_2H_6 oxidation, especially under low-to-intermediate temperatures. Therefore, it is necessary to discuss the effects of model development in this work on C_2H_5OH and C_2H_6 oxidation.

Figure 21 compares the experimental data of ignition delay times for C_2H_5OH combustion [78–81] and the corresponding simulation results by employing the developed model and other previous models [26,30,82]. All of the experimental data on ST and RCM are treated and gathered together using the method presented by Barraza-Botet et al. [80], and are scaled to a typical condition of $p = 10$ atm and $\phi = 1.0$ over a wide temperature range, using the following best-fit correlation: $\tau=7.024 \times 10^{-7} (\text{Inert}/O_2)^{0.68} p^{-0.86} \exp(15711/T)$. As shown in Fig. 21, the simulation results of the developed model on RCM data are slightly shorter than AramcoMech2.0 [30] in low temperatures, while the differences at high temperatures can be ignored. As for UCSD [26] and Hashemi et al. [82], the predictions of ignition delay times are obviously longer than the developed model. When coupled with a pressure-rise of 2.5%/ms observed by Lee et al. [81], the developed model captures the data of ST, while the traditional simulation captures the data of RCM.

A simulation of concentration profiles compared with experimental data from Dagaut and Togbe [83] in JSR is shown in Fig. 22. Compared with ignition delay times, the differences of

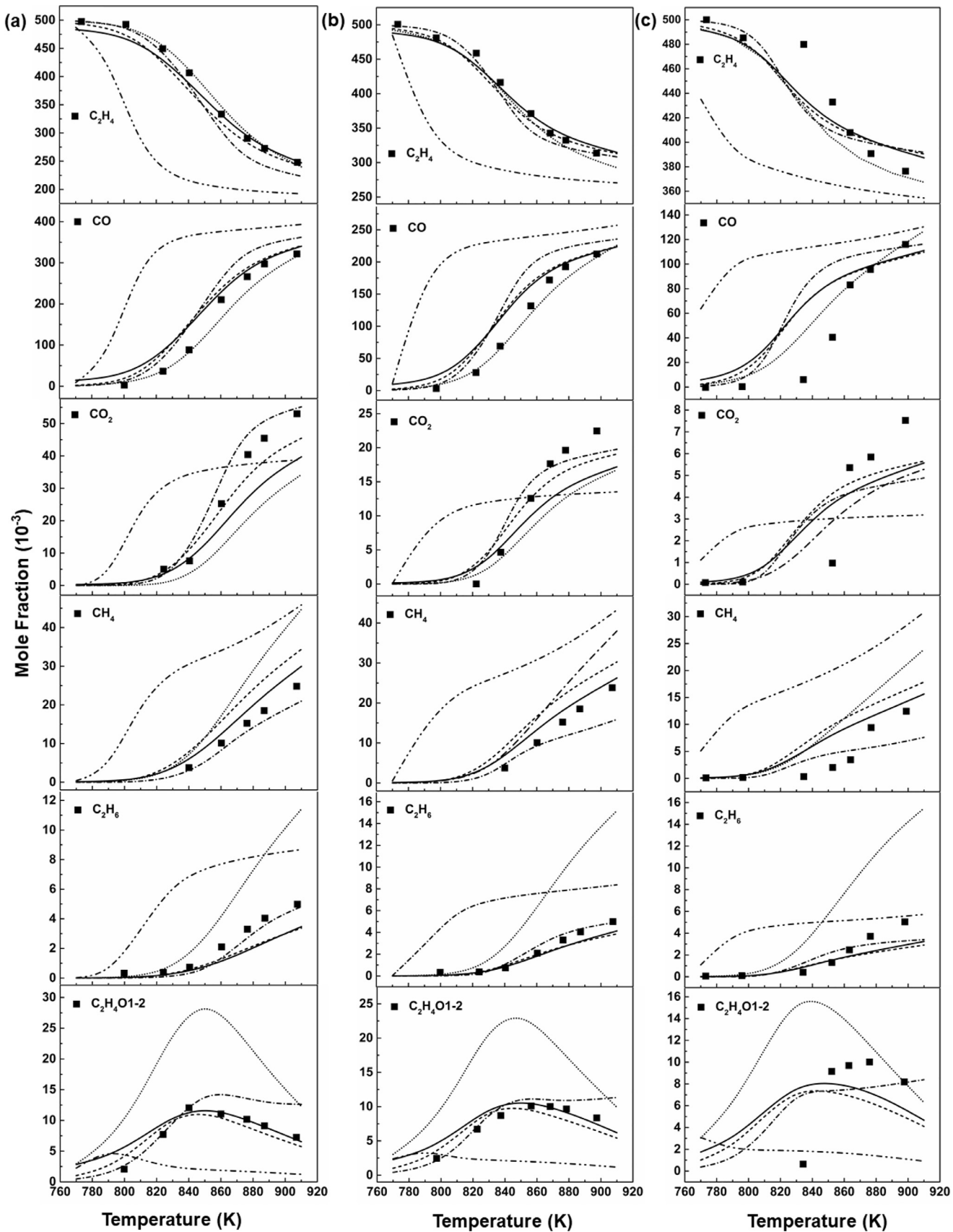


Fig. 16. Species concentration of ethylene oxidation for $C_2H_4/O_2/N_2$ mixtures at 1 atm and different equivalence ratios in JSR. The mean residence time is 1.3 s. The initial concentration of ethylene is fixed as 0.05 (mole fraction). (a) $\phi = 3.0$, 0.05 $C_2H_4/0.05 O_2/0.9 N_2$; (b) $\phi = 5.0$, 0.05 $C_2H_4/0.03 O_2/0.92 N_2$; (c) $\phi = 10.0$, 0.05 $C_2H_4/0.015 O_2/0.935 N_2$. Symbols: JSR data from Jallais et al. [39] — the developed model; - - - AramcoMech2.0 [30]; Glarborg et al. [27]; - · - UCSD [26]; - - - USC version II [18].

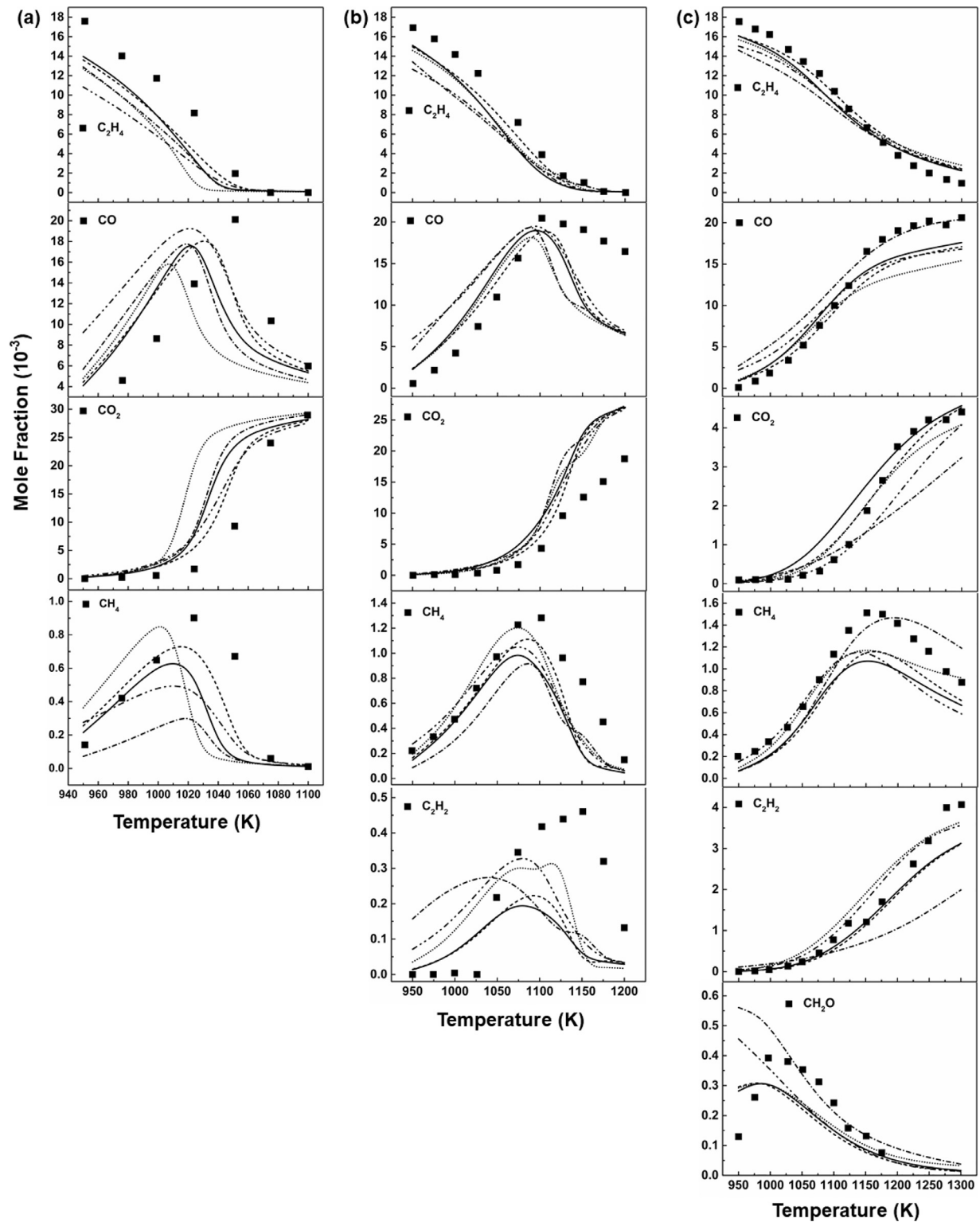


Fig. 17. Species concentration of ethylene oxidation for $C_2H_4/O_2/N_2$ mixtures at 1 atm and different equivalence ratios in JSR. The mean residence time is 0.12 s. The initial concentration of ethylene is fixed as 0.0017 (mole fraction). (a) $\phi = 0.5$, 0.0017 $C_2H_4/0.0102 O_2/0.9881 N_2$; (b) $\phi = 1.0$, 0.0017 $C_2H_4/0.0051 O_2/0.9932 N_2$; (c) $\phi = 2.0$, 0.0017 $C_2H_4/0.00255 O_2/0.99575 N_2$. Symbols: JSR data from the work of Cong et al. [38], in which the experimental data of C_2H_2 at $\phi = 0.5$, and the data of CH_2O at $\phi = 0.5$ and $\phi = 1.0$ are not available. — the developed model; - - ramcoMech2.0 [30]; Glarborg et al. [27]; -.- UCSD [26]; -.- USC version II [18].

concentration profiles using different models are more obvious in intermediate-low temperatures. It can be observed that present modifications improve the simulation results of C_2H_5OH , such as the concentration curves of C_2H_5OH , CO, C_2H_4 , O_2 , CO_2 , and C_2H_6 . These influences attribute to the detailed updates of β -

hydroxyethyl ($\dot{P}C_2H_4OH$) and α -hydroxyethyl ($\dot{S}C_2H_4OH$) chemistry in this work, as illustrated by flux analysis in Fig. 25(a). The formation of $\dot{P}C_2H_4OH$ accounts for 16.9% of total flux on C_2H_5OH consumption, and the formation of $\dot{S}C_2H_4OH$ accounts for 79%, both through H-abstraction reactions. Subsequently, about 70% of

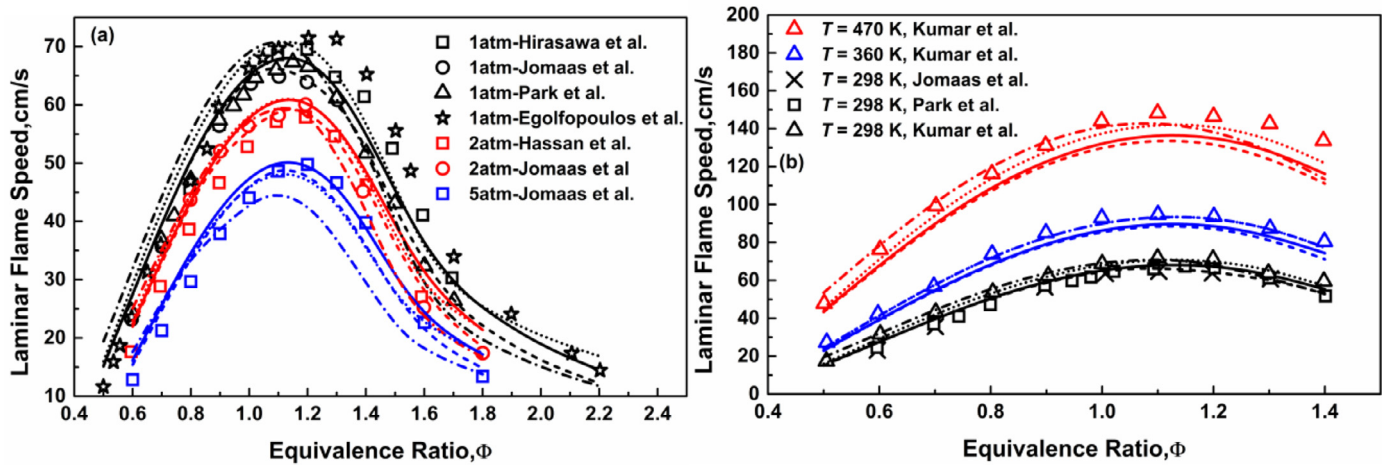


Fig. 18. Laminar flame speed for ethylene/air mixtures. (a) Effect of pressures at a fixed temperature of 298 K; (b) Effect of temperatures at a fixed pressure of 1 atm. Symbols: experimental data from the literature [22,41–45], as presented in Table 6. — the developed model; - - - AramcoMech2.0 [30]; UCSD [26]; --- Glarborg et al. [27].

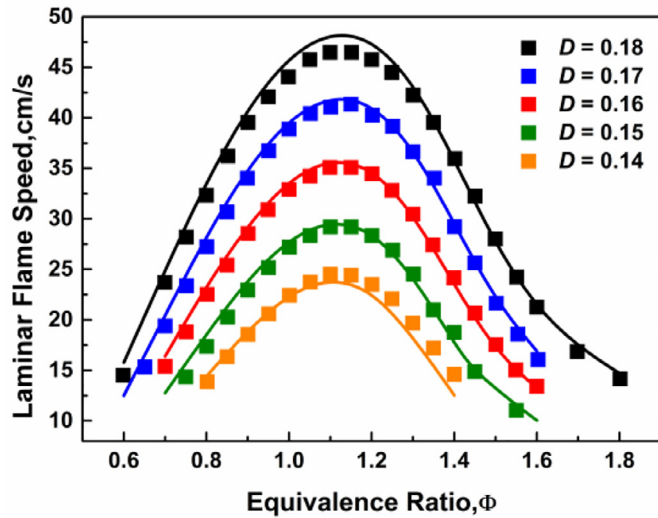


Fig. 19. Laminar flame speed for ethylene oxidation with different dilution ratios, at $T = 298$ K and $p = 1$ atm. $D = O_2/(O_2 + N_2) \times 100\%$. Symbols: experimental data from the literature [40]. Lines: simulation results.

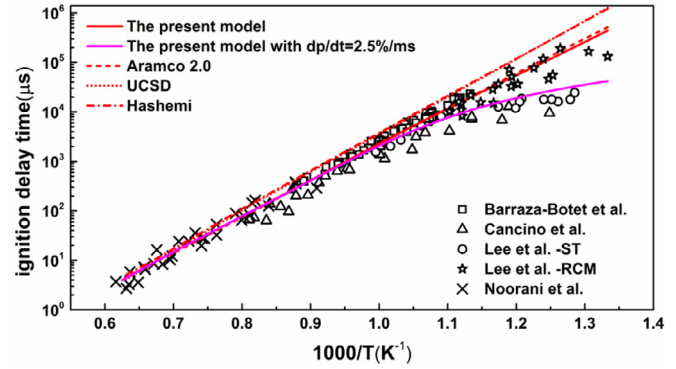


Fig. 21. The normalized ignition delay times for ethanol/air mixtures at $p = 10$ atm and $\phi = 1.0$ over a wide range of temperatures. Symbols: experimental data of Barraza-Botet et al. [80], Cancino et al. [78], Lee et al. [81] and Noorani et al. [79]. Lines: simulation results employing the developed model, AramcoMech2.0 [30], UCSD [26] and Hashemi et al. [82].

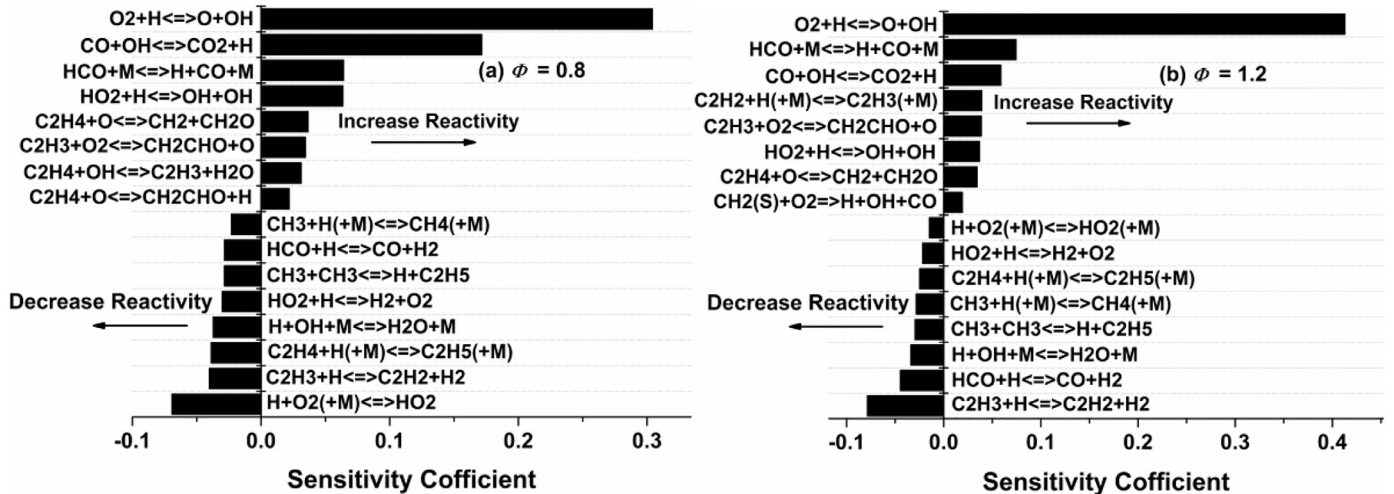


Fig. 20. Sensitivity analyses of the laminar flame speed for ethylene/air mixtures at 298 K, $p = 1$ atm, (a) $\phi = 0.8$ and (b) $\phi = 1.2$.

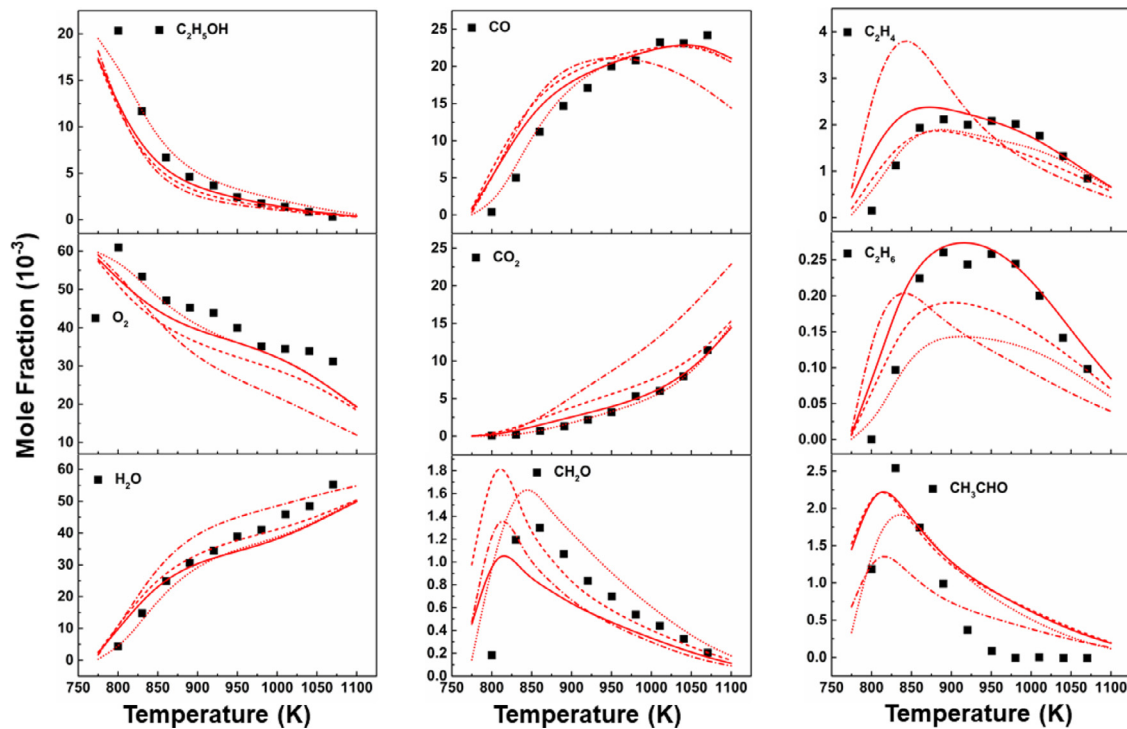


Fig. 22. Species concentration of ethanol oxidation at $\phi = 1.0$ and $p = 10.0$ atm in JSR. The mean residence time is 0.7 s, and the initial reactants are 0.002 $C_2H_5OH/0.006 O_2/0.992 N_2$ (mole fraction). Symbols: experimental from Dagaut et al. [83] — the developed model; - - - AramcoMech2.0 [30]; Hashemi et al. [82]; -·- UCSD [26].

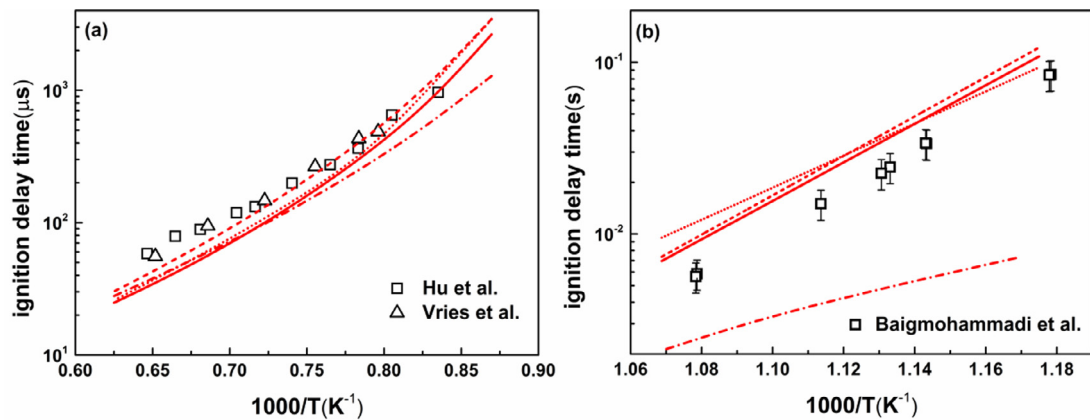


Fig. 23. Ignition delay times for ethane oxidation. (a) 0.02 $C_2H_6/0.07 O_2/0.91 Ar$ mixtures (mole fraction) in high temperatures at $\phi = 1.0$, $p = 2$ atm. Symbols: experimental data from Hu et al. [85] and Vries et al. [86] — the developed model; - - - AramcoMech2.0 [30]; UCSD [26]; -·- Hashemi et al. [84] (b) C_2H_6/air mixtures in low-to-intermediate temperatures at $\phi = 1.0$, $p = 80$ bar. Symbols: experimental data from Baigmohammadi et al. [23] — the developed model; - - - AramcoMech2.0 [30]; Glarborg et al. [27]; -·- UCSD [26].

PC_2H_4OH and almost 99% of SC_2H_4OH consumes via the oxidation reaction.

As for ethane oxidation, the study for real fuel/air mixtures and low-temperature combustion is very limited. Figure 23 exemplifies the predictions of ignition delay times by employing the present model and other popular models from literature [26,27,30,84]. The available experimental data for low-temperature oxidation of ethane are taken from the recent work of Curran's group [23], and the high-temperature data are obtained from Hu et al. [85] and Vries et al. [86]. Compared with original AramcoMech2.0, present revisions slightly shorten the ignition delay time at high temperatures, and this may due to the updated reaction class of $C_2H_4 + \dot{O}$. As shown in Fig. 23(b), the developed model slightly promotes the predictions on low-temperature ignition data of RCM, and UCSD obviously under-predicts the experimental results. Figure 24 gives the comparisons of simulated concentration profiles of different

models and experimental data in JSR from Dagaut et al. [87]. Except some differences are observed for Hashemi et al. [84], other modeling results by employing different models are close. As flux analysis shows in Fig. 25(b), the reaction forming C_2H_4 account for 91.0% on the consumption of $\dot{C}_2H_5$, reaction forming $C_2H_5\dot{O}$ (ethoxy) accounts for 5.0%, and the flux ratio for the well-skipping reaction forming $C_2H_4O_{1-2}$ is only 1.0%. However, the traditional low-temperature channels initiating from $\dot{C}_2H_5 + O_2 = C_2H_5OO\dot{O}$ (R2–3) are found not dominant. This phenomenon is not surprising because the barrier height of isomerization from $C_2H_5OO\dot{O}$ to $\dot{C}_2H_4OOH$ (R2–5) in $\dot{C}_2H_5 + O_2$ PES is above the reactants in energy, which may lead to the low-temperature chemistry inhibited, as noted in previous theoretical studies [63,88]. The slight influence of the present model on ethane predictions should be attributed to the revised or added ethylene chemistry, because a large amount of ethylene is formed in the process of ethane oxidation.

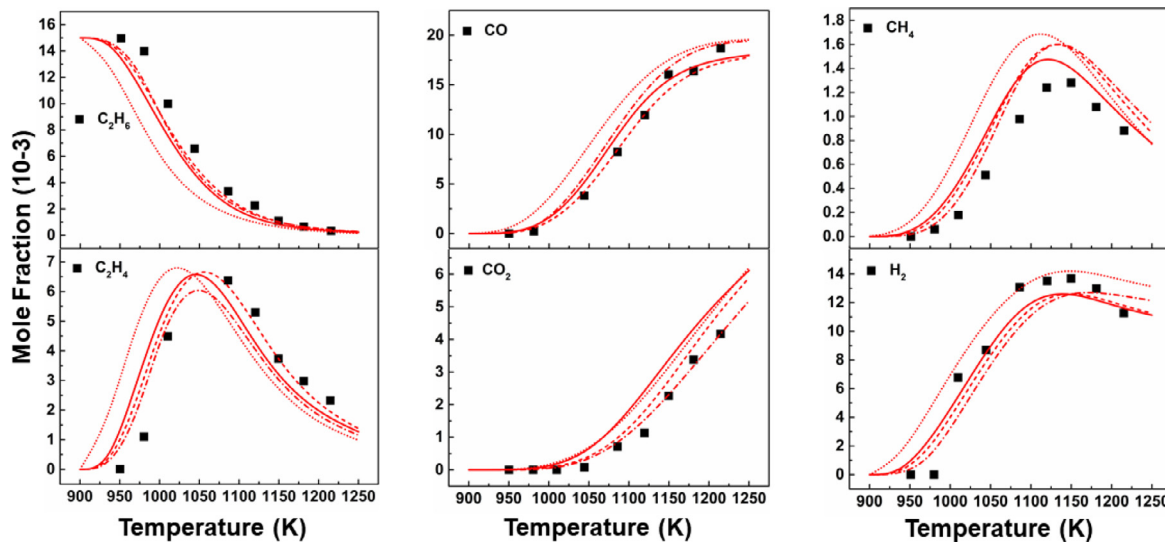


Fig. 24. Species concentration of ethane oxidation at $\phi = 1.5$ and $p = 1.0$ atm in JSR. The mean residence time is 0.1 s, and the initial reactants are 0.0015 C_2H_6 /0.0035 O_2 /0.995 N_2 (mole fraction). Symbols: experimental from Dagaut et al. [87] — the developed model; - - - AramcoMech2.0 [30]; Hashemi et al. [84]; - · - UCSD [26].

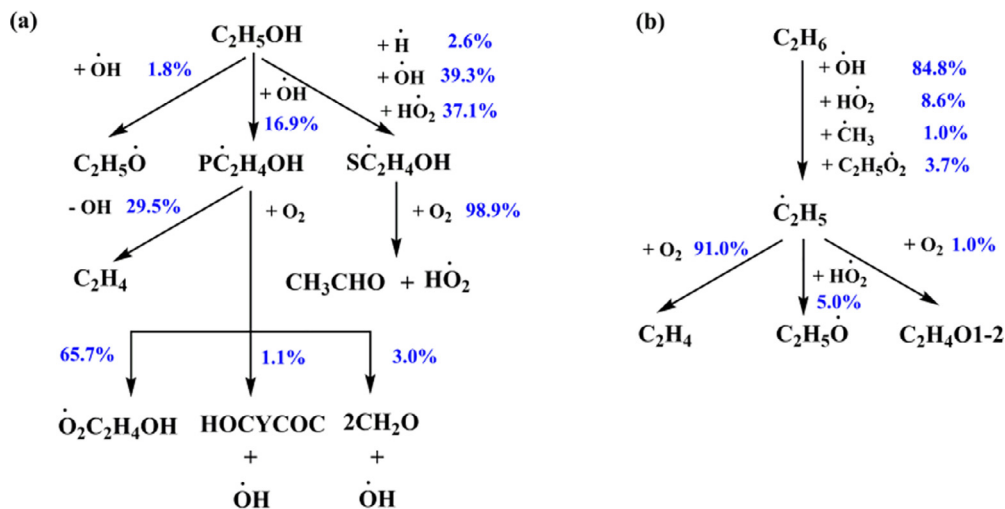


Fig. 25. Flux analysis for 20% fuel consumption using the developed model. (a) Ethanol/air mixtures at $p = 10$ atm, $\phi = 1.0$ and $T = 750$ K. (b) Ethane/air mixtures at $p = 10$ atm, $\phi = 1.0$ and $T = 856$ K.

5. Conclusions

This work presents systematic shock tube measurements of ignition delay times for ethylene/air mixtures over a wide range of temperatures (721–1320 K), pressures (1, 4, 10, 19 atm) and equivalence ratios (0.5, 1, 2). A detailed model based on AramcoMech2.0 is developed to describe the auto-ignition behavior. Some key reaction classes are revised and updated, and sensitivity analyses are carried out to determine the dominant reactions controlling fuel reactivity. At low temperatures, β -hydroxyethyl radical chemistry is comprehensively updated, and the competition between two pathways, i.e., $\dot{O}_2C_2H_4OH = HO_2C_2H_4\dot{O}$ (R3–7) and $\dot{O}_2C_2H_4OH = HO_2\dot{C}_2H_3OH$ (R3–6), causes a change in the slope of the ignition delay curve. The activation energy of the former is raised to explain the highly multi-reference effect. The combustion behavior at high temperatures is obviously improved by updating the kinetic parameters of the addition reactions between ethylene and oxygen atom ($C_2H_4 + \dot{O}$). At intermediate-to-high temperatures, the reaction class of the vinyl radical plus oxygen ($\dot{C}_2H_3 + O_2$) exerts a great impact on reactivity. The whole PES of $C_2H_4 + HO_2$ and $C_2H_5 + O_2$ is important over a wide temperature range, especially

for the well-skipping reaction $C_2H_4 + HO_2 = C_2H_4O + 2\dot{O}H$ (R2–7). The second O_2 addition reaction and subsequent dissociation reactions are also added into the model.

The current model shows a good performance on ignition delay times over a wide temperature range not only in air atmosphere but also in Ar-diluted conditions. Moreover, the developed model is extensively validated by JSR species measurements and laminar flame speed at various conditions. Great agreement is achieved for laminar flame speed, and the developed model is also able to capture the overall trend of concentration profiles for stable species. However, some discrepancies compared against experimental measurements are found for modeling predictions of concentration profiles of CO, CO_2 , CH_4 and C_2H_2 at high temperatures by employing both the present model and other core mechanisms from literature. Besides C_0 – C_1 reactions from the sub-mechanism, some C_2 chemistry, such as $\dot{C}_2H_3 + O_2$, $C_2H_4 + \dot{O}$ and $\dot{C}_2H_5 + O_2$ also influence laminar premixed flame propagation. Present model modifications influence the predictions of low-temperature combustion for both ethane and ethanol. Overall, slightly better modeling results on ignition delay times and concentration curves for some species are achieved. The model developed in this work

provides a better understanding for combustion chemistry of ethylene and serves as a cornerstone for the development of models of larger hydrocarbon fuels.

Declaration of Competing Interest

The authors declare that they have no conflicts of interest to this work that could have appeared to influence the publication of this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2020.07.017.

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